# STRUCTURED PRECONDITIONING IN AFFINE-INVARIANT GEOMETRY: PROJECTION, CERTIFICATES, AND KRONECKER SEPARATION

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**Abstract.** Nearest structured approximation and best structured preconditioning solve different matrix optimization problems. We determine their exact relation for Kronecker positive-definite matrices under the affine-invariant Riemannian metric. The Kronecker family is closed and geodesically convex, so every full matrix has a unique affine-invariant projection. Its logarithmic residual satisfies partial-trace normal equations and yields certified point and objective errors for an Armijo projection solver. Our central result shows that this unique projection is also a minimizer of the Hessian-relative condition number if and only if the extreme spectral states admit identical tensor marginals. A computable marginal-mismatch residual either vanishes at a condition-optimal projection or produces a strict descent direction. Two relative spectral levels always force projection optimality; more strongly, every $2 \times 2$ Kronecker projection is condition-optimal. An explicit $2 \times 3$ construction is therefore a dimension-minimal strict separation. Residual-calibrated bounds further bracket the best attainable Kronecker condition number and the suboptimality of the projection. Supporting results place classical diagonal and block Loewner sandwiches, fixed-basis primal–dual obstructions, and general log-spectral targets in the same certificate language. Given validated numerical enclosures and outward-rounded comparisons, an interval-safe corollary preserves the soundness of the full Kronecker tests. Deterministic small-matrix checks, including a multistart generic log-factor oracle independent of the partial-trace solver, verify the stated identities and bounds.

**Key words.** positive-definite matrix, affine-invariant metric, Kronecker product, structured preconditioning, condition number, geodesic convexity

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## 1. Introduction.

Positive-definite preconditioners define the geometry in which an optimization method converts gradients into motion. A full matrix can reshape every direction, whereas practical methods often restrict the geometry to diagonal, block, Kronecker, or low-rank structure. The restriction creates two matrix problems. The first asks whether the family can reach a prescribed Hessian-relative condition number. The second asks which structured geometry is nearest to a given full matrix. These objectives are related, but they need not select the same matrix.

Let $H \in \mathbb{S}^d_{++}$ and let $G \in \mathbb{S}^d_{++}$ be a candidate geometry. The preconditioned condition number is

$$\kappa(G^{-1}H) = \kappa(H^{-1/2}GH^{-1/2}).$$

We measure matrix displacement with the affine-invariant distance

$$d_{\mathrm{AI}}(G_0, G_1) = \left\| \log(G_0^{-1/2} G_1 G_0^{-1/2}) \right\|_F.$$

This metric is congruence invariant and turns the SPD cone into a Hadamard manifold. It therefore supplies unique projection onto closed geodesically convex sets and a natural intrinsic geometry for structured matrix families.

Our first layer treats spectral targets. For $H_a \succ 0$, let

$$\Omega_a(G) = \phi_a\Big(\log \lambda(G^{-1/2} H_a G^{-1/2})\Big),$$

where $\phi_a$ is convex and permutation invariant. Coordinatewise nondecreasing convex aggregation preserves geodesic convexity. This class contains log condition number and

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smooth spectral surrogates. Its sublevel sets have unique affine-invariant projection, an explicit exact penalty under a Slater margin, and a proximal iteration with a global contraction bound. For closed geodesically convex smooth families, optimality is equivalent to the whitened spectral subgradient lying in the normal space.

The second layer gives family-specific certificates. Diagonal and fixed-block families reach a threshold $K$ exactly when a structured positive matrix $E$ satisfies

$$E \preceq H \preceq KE.$$

This is an LMI feasibility problem and admits explicit dual obstructions. For Kronecker metrics, the factorization $G = A \otimes B$ has a scale gauge: $(cA) \otimes (c^{-1}B) = A \otimes B$. After quotienting that gauge, we derive the intrinsic affine-invariant line element and distance, prove that the Kronecker set is a complete totally geodesic manifold, and characterize its nearest-point projection by partial-trace normal equations.

The projection geometry exposes the paper's central structural distinction. The nearest Kronecker point to $H$ minimizes $d_{\mathrm{AI}}(G, H)$, whereas the best Kronecker preconditioner minimizes $\kappa(G^{-1}H)$. We prove an exact criterion for when the unique projection is a condition-number minimizer, in terms of the tensor marginals of the extreme eigenspaces of the relative matrix. We also turn the criterion into a quantitative marginal-mismatch residual and a strict descent direction. Two relative spectral levels force the criterion. Independently, every $2 \times 2$ Kronecker projection is condition-optimal, while an explicit $2 \times 3$ example gives the dimension-minimal strict separation.

*Contributions..* The paper makes four main contributions.

1. We characterize exactly when the unique AIRM-nearest Kronecker matrix is also condition-optimal, express the criterion through extreme-state tensor marginals, derive a computable mismatch/descent certificate, prove projection optimality for every two-level relative spectrum and every $2 \times 2$ problem, and exhibit dimension-minimal strict separation in $2 \times 3$.
2. Building on the established Fisher/AIRM quotient geometry of Kronecker covariances, we prove global projection normal equations and give residual point-error, objective-gap, threshold, and projection-suboptimality certificates together with an Armijo solver.
3. We place these results inside a geodesically convex log-spectral target framework with a global normal-response law, exact penalty, proximal contraction, and a sharp scale-closed distance lower bound. The full Kronecker certificate admits an interval-safe form when validated enclosures of the numerical inputs are supplied.
4. As supporting comparisons, we restate classical diagonal and block reachability as Loewner-sandwich LMIs in the same target-distance notation, record dual and fixed-basis obstructions, and validate the matrix identities with deterministic, randomized, and independent-oracle checks.

The geometric objects are motivated by diagonal adaptive methods, K-FAC, Shampoo, and factored statistics, but the exact claims are deterministic finite-dimensional matrix statements. Small-SPD experiments verify projection residuals, reachability, dual obstructions, and strict separation; they are not used as optimizer-ranking evidence.

*Organization..* Section 2 reviews SPD geometry, scaling, and Kronecker approximation. Section 3 develops spectral targets and normal response. Section 4 gives diagonal and block certificates. Section 5 develops Kronecker projection and separation. Sections 6 and 7 give verification and scope.

## 2. Related Work.

*Positive-definite matrix geometry..* The affine-invariant geometry of positive-definite matrices provides the distance, exponential map, congruence invariance, and nonpositive curvature used throughout [6, 32, 17]. Geodesically convex optimization and best approximation on SPD manifolds are developed in [39, 25, 5, 41]. Spectral subdifferential calculus follows the Hermitian convex-analysis framework of [24]. Our contribution is the joint specialization to structured condition targets, explicit penalty and certificate interfaces, and the exact comparison between metric projection and condition-number optimization.

*Scaling and structured preconditioning..* Diagonal and block condition-number optimization is connected to optimal scaling, equilibration, and semidefinite formulations [4, 43, 34, 35, 27, 26, 18, 33, 13, 14]. In particular, mixed packing–covering SDP solvers and customized large-scale methods address the computational diagonal problem. Geodesically convex preconditioning over symmetric Lie-group quotients provides general condition-number convexity, projected gradients, and first-order complexity bounds [10]. We use that established convexity viewpoint as background. Applied to a Kronecker transformation group, that framework supplies condition-objective convexity and first-order optimization, but does not compare the optimizer with the AIRM projection or derive the extreme-state marginal criterion. Our diagonal and block results retain the exact Loewner reachability question in the notation used later for Kronecker target distance; the new comparison is between AIRM projection and spectral condition optimality on the Kronecker SPD submanifold.

Diagonal adaptive methods include AdaGrad and Adam [11, 20]; Adafactor and SM3 use factored or shared statistics [36, 3]; K-FAC and KFC use Kronecker curvature approximations [28, 15]; and Shampoo uses tensor-factor preconditioners [16, 2]. These algorithms motivate the matrix families. Identifying a concrete optimizer state with a point in the family additionally requires a realization theorem specifying damping, bias correction, update scale, and interpolation.

*Kronecker approximation and covariance geometry..* Kronecker approximation is classical under Frobenius loss [44]. Application-driven nearest-Kronecker preconditioners use Frobenius or operator-specific product approximations for Toeplitz, imaging, Markov/SAN, and stochastic Galerkin systems [19, 31, 22, 23, 42]. Those works ask whether a computationally convenient product approximates a structured operator well or accelerates a particular solver. Our comparison instead fixes one Kronecker family and separates its intrinsic AIRM projection from its exact Hessian-relative condition-number minimizer.

Separable covariance and matrix-normal models lead to likelihood and flip-flop procedures [12, 47]. Geodesic convexity for covariance estimation and Kronecker structure is established in [48, 49]; a broader survey is [46]. The Fisher/AIRM quotient geometry, determinant-one factorization, scale gauge, and dimension-weighted product metric are developed in [29], with related Riemannian algorithms in [7, 37]. Work connecting Shampoo factors to Kronecker approximation studies a complementary algorithmic question [30].

| Problem | Established object | Role in this paper |
|---|---|---|
| Kronecker approximation | Frobenius nearest product [44] | Contrast with intrinsic AIRM projection |
| Separable covariance | Likelihood equations and geodesic convexity [12, 49] | Distinguish statistical likelihood from metric projection |
| Kronecker information geometry | Scale quotient and product Fisher/AIRM metric [29] | Infrastructure for logarithmic partial-trace projection residuals |
| Optimal preconditioning | Geodesically convex condition optimization and first-order methods [10] | Exact criterion for when the AIRM projection belongs to the condition-number argmin |

Thus the quotient metric itself is supporting infrastructure. The main increment here is the global AIRM projection characterization, its residual error and reachability brackets, the extreme-state marginal criterion for projection optimality, and strict separation from best Hessian-relative preconditioning.

**3. Spectral Targets in Affine-Invariant Geometry.** We begin with affine-invariant SPD geometry and isolate the static spectral principles used by the structured families.

**3.1. Relative SPD geometry.** Let

$$f(\theta) = \frac{1}{2}\theta^\top H\theta, \qquad H \in \mathbb{S}^d_{++}.$$

For a metric $G \in \mathbb{S}^d_{++}$, gradient flow is $\dot{\theta} = -G^{-1}H\theta$. Define the relative metric

$$S = H^{-1/2}GH^{-1/2}.$$

Then $\kappa(G^{-1}H) = \kappa(S)$, where the left side denotes the ratio of generalized eigenvalues in $Hv = \lambda Gv$.

The affine-invariant metric has line element and distance

$$\|Z\|_{S,\mathrm{AI}} = \left\|S^{-1/2}ZS^{-1/2}\right\|_F, \qquad d_{\mathrm{AI}}(S_0, S_1) = \left\|\log(S_0^{-1/2}S_1S_0^{-1/2})\right\|_F.$$

The SPD cone is a finite-dimensional Hadamard manifold [6, 32].

For $K \geq 1$, set

$$\mathcal{C}_K = \{S \in \mathbb{S}^d_{++} : \kappa(S) \leq K\}.$$

Theorem 3.1 (Full-SPD spectral benchmark). *Let $y_1 \leq \cdots \leq y_d$ be the ordered log-eigenvalues of $S_0 \in \mathbb{S}^d_{++}$. Then*

$$D_K(S_0) := d_{\mathrm{AI}}(S_0, \mathcal{C}_K) = \min_{c\in\mathbb{R}} \left(\sum_{i=1}^d \mathrm{dist}(y_i, [c, c+\log K])^2\right)^{1/2}.$$

Thus the unconstrained benchmark clips the relative log-spectrum into an interval of width $\log K$. Restricted families constrain which spectral and eigenspace motions are admissible.

**3.2. Structured metric families and target distance.** Let $\mathfrak{F} \subset \mathbb{S}^d_{++}$ be an admissible metric family and define its Hessian-relative realization

$$\mathcal{S}_{\mathfrak{F}}(H) = \{H^{-1/2}GH^{-1/2} : G \in \mathfrak{F}\}.$$

A realization specifies a path-connected component, treatment of scale freedom, an admissible absolutely continuous path class, and the induced or quotient affine-invariant length. Smooth embedded, closed geodesically convex, quotient, and stratified families therefore remain distinct cases.

DEFINITION 3.2 (Structured affine-invariant target distance). *For $G_0 \in \mathfrak{F}$ and $S_0 = H^{-1/2} G_0 H^{-1/2}$, define*

$$D_{K,\mathfrak{F}}(S_0; H) = \inf\{\, \mathrm{Len}_{\mathrm{AI}}(S_{[0,T]}) : S(0) = S_0,\ S(T) \in \mathcal{C}_K,\\ S_t \in \mathcal{S}_{\mathfrak{F}}(H),\ S_{[0,T]}\ \textit{admissible}\}.$$

*The value is $+\infty$ if the relative target is empty or lies in a different admissible component.*

This target distance records the least intrinsic affine-invariant motion from the initial geometry to a structured condition target. The endpoint records visible preconditioning quality, and the admissible curve is eliminated by the infimum. For closed geodesically convex realizations it reduces to a unique target projection; nonconvex or stratified families retain the variational definition without inheriting that uniqueness.

**3.3. Generating geodesically convex information.** For $H_a \in \mathbb{S}^d_{++}$ and a convex permutation-invariant function $\phi_a : \mathbb{R}^d \to \mathbb{R}$, define

$$\Omega_a(G) = \phi_a\Big(\log \lambda(G^{-1/2} H_a G^{-1/2})\Big).$$

THEOREM 3.3 (Information-spectrum generation law). *Let $\mathfrak{F} \subset \mathbb{S}^d_{++}$ be nonempty, closed, and geodesically convex. If $\rho : \mathbb{R}^q \to \mathbb{R}$ is convex and nondecreasing in every coordinate, then*

$$\mathcal{J}(G) = \rho(\Omega_1(G), \ldots, \Omega_q(G))$$

*is closed and geodesically convex on $\mathfrak{F}$. The conclusion holds pointwise for continuously time-varying $H_{t,a}, \phi_{t,a}, \rho_t$.*

*If every $\phi_a(x + c\mathbf{1}) = \phi_a(x)$, then $\mathcal{J}(cG) = \mathcal{J}(G)$. If $\phi_a$ is $L_a$-Lipschitz and $\rho$ is $L_\rho$-Lipschitz, replacing $H_a$ by $\widehat{H}_a$ changes the objective by at most*

$$\sup_{G \in \mathfrak{F}} |\widehat{\mathcal{J}}(G) - \mathcal{J}(G)| \le L_\rho \left( \sum_{a=1}^{q} L_a^2 d_{\mathrm{AI}}(H_a, \widehat{H}_a)^2 \right)^{1/2}.$$

The condition-number objective is obtained from

$$\phi_{\mathrm{osc}}(x) = \max_i x_i - \min_i x_i, \qquad \mathcal{J}_H(G) = \log \kappa(G^{-1} H).$$

**3.4. Static action value, exact penalty, and contraction.** The static theorem is stated on an abstract finite-dimensional Hadamard manifold so that it can be reused by every closed geodesically convex family.

THEOREM 3.4 (Restricted target projection and exact penalty). *Let $(\mathcal{M}, d)$ be a finite-dimensional Hadamard manifold, $\mathfrak{F} \subset \mathcal{M}$ a nonempty closed geodesically convex set, and $\mathcal{J} : \mathfrak{F} \to \mathbb{R} \cup \{+\infty\}$ proper, closed, geodesically convex, and bounded below. Suppose*

$$\mathcal{C}_\tau = \{G \in \mathfrak{F} : \mathcal{J}(G) \le \tau\}$$

*is nonempty. For quadratic kinetic action and terminal indicator* $\iota_{\mathcal{C}_\tau}$*, the value is*

$$V(t,G) = \frac{d(G,\mathcal{C}_\tau)^2}{2(T-t)}.$$

*For* $G_0 \in \mathfrak{F}$*, the least admissible length is*

$$\mathfrak{D}^{\mathcal{J}}_{\tau,\mathfrak{F}}(G_0) = d(G_0,\mathcal{C}_\tau) = d(G_0,P_\tau), \qquad P_\tau = P_{\mathcal{C}_\tau}(G_0).$$

*The projection and constant-speed minimizing geodesic are unique, and for every* $Y \in \mathcal{C}_\tau$*,*

$$d(G_0,Y)^2 \geq d(G_0,P_\tau)^2 + d(P_\tau,Y)^2.$$

*If families or thresholds are nested, their distances are monotone. If the objectives satisfy*

$$\sup_{G\in\mathfrak{F}} |\widehat{\mathcal{J}}(G) - \mathcal{J}(G)| \leq \varepsilon,$$

*then*

$$\mathfrak{D}^{\mathcal{J}}_{\tau+\varepsilon,\mathfrak{F}}(G_0) \leq \mathfrak{D}^{\widehat{\mathcal{J}}}_{\tau,\mathfrak{F}}(G_0) \leq \mathfrak{D}^{\mathcal{J}}_{\tau-\varepsilon,\mathfrak{F}}(G_0)$$

*whenever the outer values are finite.*

*Assume a strict feasible geometry* $G_s$ *satisfies* $\mathcal{J}(G_s) \leq \tau - \sigma$*,* $\sigma > 0$*, and put* $R = d(G_0,G_s) > 0$*. On* $\mathcal{D} = \mathfrak{F} \cap \overline{B}_R(G_0)$*, every*

$$\Lambda > \frac{2R^2}{\sigma}$$

*makes*

$$\Phi_\Lambda(G) = \frac{1}{2} d(G_0,G)^2 + \Lambda\,[\mathcal{J}(G) - \tau]_+$$

*an exact penalty with unique minimizer* $P_\tau$*. For every* $\eta > 0$*, the proximal iteration*

$$G_{r+1} = \operatorname*{arg\,min}_{G\in\mathcal{D}} \left\{ \Phi_\Lambda(G) + \frac{d(G,G_r)^2}{2\eta} \right\}$$

*satisfies*

$$d(G_r,P_\tau)^2 \leq (1+2\eta)^{-r} d(G_0,P_\tau)^2.$$

*If* $R = 0$*, then* $G_0 = P_\tau$*.*

The proximal recursion is a variational convergence benchmark: its rate is conditional on solving each global proximal subproblem. No claim is made that those subproblems are cheaper than the original target projection for an arbitrary family.

For the condition-number target, take $\mathcal{J} = \mathcal{J}_H$ and $\tau = \log K$. Then $\mathfrak{D}^{\mathcal{J}}_{\tau,\mathfrak{F}} = D_{K,\mathfrak{F}}$ for every closed geodesically convex realization.

**3.5. Normal response and the nearest-versus-best question.** Let $\mathfrak{F}$ now be a nonempty, closed, geodesically convex smooth embedded submanifold. In whitened coordinates define

$$\widehat{T}_G\mathfrak{F} = \{G^{-1/2} U G^{-1/2} : U \in T_G\mathfrak{F}\}, \qquad \widehat{N}_G\mathfrak{F} = (\widehat{T}_G\mathfrak{F})^\perp.$$

For $S_a = G^{-1/2} H_a G^{-1/2}$, let

$$\mathfrak{Z}_a(G) = \partial_X[\phi_a(\lambda(X))]_{X=\log S_a}.$$

Here $\partial_X$ is the Euclidean convex subdifferential in the whitened symmetric-matrix coordinate $X$; the affine-invariant tangent/cotangent identification is applied only after this whitening.

LEMMA 3.5 (Log-spectral pullback identity). *Let $S \in \mathbb{S}^d_{++}$, let $\phi : \mathbb{R}^d \to \mathbb{R}$ be convex and permutation invariant, and set $F(Y) = \phi(\lambda(Y))$ on $\mathbb{S}^d$. For every $X \in \mathbb{S}^d$,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0+} F\Big(\log(e^{-tX/2} S e^{-tX/2})\Big) = \max_{Z \in \partial F(\log S)} -\operatorname{tr}(ZX).$$

*Consequently, the whitened AIRM subdifferential of the corresponding relative log-spectral objective is exactly $-\partial F(\log S)$, including at repeated eigenvalues.*

THEOREM 3.6 (Normal-response law). *A point $G \in \mathfrak{F}$ globally minimizes the information objective in Theorem 3.3 if and only if there exist*

$$\theta \in \partial\rho(\Omega_1(G), \ldots, \Omega_q(G)), \qquad Z_a \in \mathfrak{Z}_a(G),$$

*such that*

$$\sum_{a=1}^{q} \theta_a Z_a \in \widehat{N}_G \mathfrak{F}.$$

*Thus optimality is exact cancellation of the tangent information force.*

*For a single $H$, let $P = P_{\mathfrak{F}}(H)$ be its unique nearest affine-invariant point in $\mathfrak{F}$ and set $S = P^{-1/2} H P^{-1/2}$. Then $\log S \in \widehat{N}_P \mathfrak{F}$, and $P$ also minimizes the log-spectral difficulty $\Omega_{\phi,H}$ if and only if*

$$\mathfrak{Z}_\phi(P) \cap \widehat{N}_P \mathfrak{F} \neq \emptyset.$$

COROLLARY 3.7 (Sharp scale-closed threshold). *Let $d \geq 2$ and let $\mathfrak{F} \subset \mathbb{S}^d_{++}$ be nonempty and closed under positive rescaling. Define*

$$K^\star_{\mathfrak{F}}(H) = \inf_{G \in \mathfrak{F}} \kappa(G^{-1}H), \qquad \delta_{\mathfrak{F}}(H) = d_{\mathrm{AI}}(H, \mathfrak{F}), \qquad \alpha_d = \sqrt{\frac{d}{\lfloor d^2/4 \rfloor}}.$$

*Then*

$$\log K^\star_{\mathfrak{F}}(H) \geq \alpha_d \delta_{\mathfrak{F}}(H), \qquad K^\star_{\mathfrak{F}}(H) \geq \exp\{\alpha_d d_{\mathrm{AI}}(H, \mathfrak{F})\}.$$

*The constant $\alpha_d$ is sharp over general scale-closed families.*

PROPOSITION 3.8 (Monotonicity and full-SPD lower bound). *If $\mathfrak{F}_1 \subset \mathfrak{F}_2$ and every admissible $\mathfrak{F}_1$-path is an admissible $\mathfrak{F}_2$-path with the same length, then*

$$D_{K,\mathfrak{F}_1}(S_0; H) \geq D_{K,\mathfrak{F}_2}(S_0; H).$$

*In particular, $D_{K,\mathfrak{F}}(S_0; H) \geq D_K(S_0)$ whenever restricted paths are ambient SPD paths. Also $D_{K_1,\mathfrak{F}} \geq D_{K_2,\mathfrak{F}}$ for $K_1 \leq K_2$.*

PROPOSITION 3.9 (Intrinsic submanifold distance). *If $\mathcal{S}_{\mathfrak{F}}(H)$ is a smooth embedded submanifold equipped with the ambient induced length, then*

$$D_{K,\mathfrak{F}}(S_0; H) = d_{\mathcal{S}_{\mathfrak{F}}(H)}(S_0, \mathcal{S}_{\mathfrak{F}}(H) \cap \mathcal{C}_K).$$

*The distance is $+\infty$ when the target lies outside the admissible path-connected component.*

**4. Diagonal and Block Matrix Families.** Diagonal and block families are the closest finite-dimensional models of coordinatewise and grouped adaptive preconditioning. Their reachability admits a clean semidefinite characterization. This connects restricted geometric complexity to the classical literature on scaling and equilibration [4, 43, 34, 35, 38, 21, 33], to condition-number optimization through convex programming [27, 26], and to semidefinite feasibility [45, 9]. The role of this section is to place those reachability questions inside the structured affine-invariant target-distance benchmark and to record primal and dual certificates in the notation used by $D_{K,\mathfrak{F}}$.

| Topic | Closest classical object | Role here |
|---|---|---|
| Diagonal/block scaling | Optimal scaling and equilibration criteria | Endpoint reachability for a restricted metric family |
| Condition-number minimization | Variational and convex-programming formulations | The same target embedded in an affine-invariant path-distance benchmark |
| Semidefinite feasibility | Primal feasibility and dual separation | Certificate language for finite $D_{K,\mathfrak{F}}$ or unreachable targets |

**4.1. Diagonal reachability.** Let

$$\mathfrak{F}_{\mathrm{diag}} = \{D = \mathrm{diag}(e^{x_1}, \dots, e^{x_d}) : x \in \mathbb{R}^d\}.$$

Inside this family the affine-invariant line element is Euclidean in log coordinates:

$$\mathrm{Len}_{\mathrm{diag}}(x_{[0,T]}) = \int_0^T \|\dot{x}_t\|_2 \,\mathrm{d}t.$$

The diagonal target set is

$$\mathcal{X}_K(H) = \{x \in \mathbb{R}^d : \kappa(\mathrm{diag}(e^{-x})H) \le K\}.$$

LEMMA 4.1 (Diagonal distance identity). *The map*

$$x \mapsto H^{-1/2}\,\mathrm{diag}(e^{x})H^{-1/2}$$

*is an isometric realization of the diagonal metric family with its affine-invariant induced length. With $S_0 = H^{-1/2}\,\mathrm{diag}(e^{x_0})H^{-1/2}$, write $D_{K,\mathrm{diag}}(x_0; H) := D_{K,\mathrm{diag}}(S_0; H)$. The set $\mathcal{X}_K(H)$ is closed, and*

$$D_{K,\mathrm{diag}}(x_0; H) = \mathrm{dist}_{\ell^2}(x_0, \mathcal{X}_K(H)).$$

THEOREM 4.2 (Diagonal LMI reachability). *For $K \ge 1$, the following are equivalent:*
1. *there exists a positive diagonal $D$ with $\kappa(D^{-1}H) \le K$;*
2. *there exists a positive diagonal $E$ with*

$$E \preceq H \preceq KE.$$

*Consequently,*

$$K^*_{\mathrm{diag}}(H) = \inf\{K \ge 1 : \exists E \succ 0 \text{ diagonal},\ E \preceq H \preceq KE\}.$$

*For fixed $K$, diagonal reachability is an LMI feasibility problem.*

*Remark* 4.3 (Certificate interpretation). Theorem 4.2 separates reachability from path length. If the LMI is infeasible, then $D_{K,\mathrm{diag}}(x_0; H) = +\infty$ for every initial diagonal metric. If it is feasible, the remaining problem is the intrinsic distance from the initial log-scale $x_0$ to the feasible set $\mathcal{X}_K(H)$. Semidefinite duality can therefore provide certificates that no diagonal preconditioner can reach a desired condition-number threshold.

COROLLARY 4.4 (Aligned diagonal optimum). *If $H = \mathrm{diag}(h_1, \ldots, h_d)$ and $G_0$ is diagonal, then*

$$D_{K,\mathrm{diag}}(S_0; H) = D_K(S_0).$$

Thus diagonal geometry can be optimal when the target curvature is already aligned with the coordinate axes. Its loss comes from directional mismatch.

**4.2. Block reachability.** Let the coordinates be split as

$$\mathbb{R}^d = V_1 \oplus \cdots \oplus V_m, \qquad \dim V_j = d_j,$$

and let $\mathfrak{F}_{\mathrm{block}}$ be the corresponding block-diagonal positive definite family.

THEOREM 4.5 (Block LMI reachability). *For $K \geq 1$, the following are equivalent:*
1. *there exists $G \in \mathfrak{F}_{\mathrm{block}}$ with $\kappa(G^{-1}H) \leq K$;*
2. *there exists $E \in \mathfrak{F}_{\mathrm{block}}$ with*

$$E \preceq H \preceq KE.$$

*Therefore block reachability is also an LMI feasibility problem.*

**4.3. Dual infeasibility certificates.** The LMI formulation also gives a checkable certificate when a target condition number is impossible. Let $\mathcal{L}$ be either the diagonal symmetric subspace or a fixed block-diagonal symmetric subspace, and let $\Pi_{\mathcal{L}}$ be the Frobenius-orthogonal projection onto $\mathcal{L}$. For fixed $K$, the primal reachability problem asks for $E \in \mathcal{L}$ such that

$$H - E \succeq 0, \qquad KE - H \succeq 0.$$

PROPOSITION 4.6 (Structured SDP dual certificate). *If there exist $P, Q \in \mathbb{S}^d_+$ such that*

$$\Pi_{\mathcal{L}}(P - KQ) = 0, \qquad \langle P - Q, H \rangle < 0,$$

*then no $E \in \mathcal{L}$ satisfies $E \preceq H \preceq KE$. Hence the diagonal or block restricted target is empty at threshold $K$.*

COROLLARY 4.7 (Sign-flip rank-one certificate). *Let*

$$T = \mathrm{blockdiag}(s_1 I_{d_1}, \ldots, s_m I_{d_m}), \qquad s_j \in \{-1, 1\},$$

*and let $\mathcal{L}$ be the corresponding block-diagonal subspace. If*

$$\lambda_{\min}(KTHT - H) < 0,$$

*then the block family cannot reach condition number $K$. The diagonal case is the special case $d_j = 1$.*

The sign-flip certificate is not meant to solve every infeasible SDP. It gives a cheap, inspectable dual witness for synthetic examples where the obstruction is cross-block or cross-coordinate coupling.

Proposition 4.8 (Block Jacobi upper bound). *Let*

$$G_{\mathrm{BJ}} = \operatorname{blockdiag}(H_{11}, \ldots, H_{mm})$$

*with every* $H_{jj} \succ 0$, *and set*

$$\widetilde{H} = G_{\mathrm{BJ}}^{-1/2} H G_{\mathrm{BJ}}^{-1/2}.$$

*Then*

$$K^*_{\mathrm{block}}(H) \le \kappa(\widetilde{H}).$$

*If* $\left\|\widetilde{H} - I\right\|_{\mathrm{op}} \le \delta < 1$, *then*

$$K^*_{\mathrm{block}}(H) \le \frac{1+\delta}{1-\delta}.$$

The block family can absorb all within-block curvature. The residual difficulty is the normalized off-block interaction.

**5. Kronecker Geometry and Reachability.** Matrix and tensor parameters motivate Kronecker and low-rank restrictions. These families require more care than diagonal or block families because their parameterizations include gauge redundancies and nonconvex structure. They model the structural assumptions behind Kronecker-factored natural gradient and tensor preconditioning methods [28, 16]. They are also related to classical Kronecker approximation and separable covariance estimation [44, 12, 47]; the objective here is affine-invariant metric geometry rather than Frobenius approximation or statistical likelihood. This section proves a local quotient line element, a fixed-basis spectral equivalence, and a general affine-invariant projection theorem for Kronecker metrics. The projection theorem does not give an elementary closed form in the fully noncommuting case, but it gives a unique target, partial-trace normal equations, certified residuals, and auxiliary self-conditioned $K$-target bounds. The Hessian-relative target has an exact endpoint reachability formulation as a nonconvex Kronecker Loewner sandwich, an associated expression threshold, residual-calibrated threshold brackets, and an exact fixed-basis primal–dual obstruction theorem; candidate Kronecker metrics give direct upper bounds on $D_{K,\mathcal{K}}$ whenever they pass the generalized condition-number test against $H$. The intended output is a certifying interface: a factor-path line element, spectral mismatch signals, fixed-basis primal or dual witnesses, and noncommuting Kronecker certificates that prove reachability, prove some global impossibility cases, or abstain.

**5.1. Kronecker spectral surrogate.** Let $W \in \mathbb{R}^{m\times n}$, so that $\operatorname{vec}(W) \in \mathbb{R}^{mn}$. A Kronecker metric has the form

$$G = B \otimes A, \qquad A \in \mathbb{S}^m_{++}, \quad B \in \mathbb{S}^n_{++}.$$

If

$$A = U \operatorname{diag}(a_i) U^\top, \qquad B = V \operatorname{diag}(b_j) V^\top,$$

then $B \otimes A$ has eigenvectors $v_j \otimes u_i$ and eigenvalues $b_j a_i$. Its log-spectrum has the additive form

$$\log a_i + \log b_j.$$

This motivates the additive subspace

$$\mathcal{A}_{\mathrm{kron}} = \{Z \in \mathbb{R}^{m\times n} : Z_{ij} = \alpha_i + \beta_j\}.$$

*Remark* 5.1 (Status of the spectral surrogate). The additive log-spectrum model below is a diagnostic subproblem. It becomes an intrinsic Kronecker complexity only in the shared fixed-eigenbasis regime of Proposition 5.6. Outside that regime, Kronecker structure also constrains eigenvectors; the quotient geometry in Proposition 5.4 and the projection theorem below are the geometric objects.

In a fixed shared Kronecker eigenbasis, let $Y \in \mathbb{R}^{m\times n}$ denote the current relative log-spectrum. A Kronecker spectral update can move inside the affine set $Y + \mathcal{A}_{\mathrm{kron}}$. Define

$$\mathcal{Y}_K = \{Z : \max_{i,j} Z_{ij} - \min_{i,j} Z_{ij} \le \log K\}.$$

DEFINITION 5.2 (Kronecker spectral complexity). *The Kronecker spectral surrogate is*

$$D^{\mathrm{spec}}_{K,\mathrm{kron}}(Y) = \mathrm{dist}_F\big(Y,\ (Y + \mathcal{A}_{\mathrm{kron}}) \cap \mathcal{Y}_K\big),$$

*with value $+\infty$ if the intersection is empty.*

PROPOSITION 5.3 (Spectral lower bound). *Let $D_K(Y) = \mathrm{dist}_F(Y, \mathcal{Y}_K)$. Then*

$$D^{\mathrm{spec}}_{K,\mathrm{kron}}(Y) \ge D_K(Y).$$

*If the full projection $\Pi_{\mathcal{Y}_K}(Y)$ lies in $(Y + \mathcal{A}_{\mathrm{kron}})$, then equality holds.*

This surrogate is useful but incomplete: the true Kronecker family also restricts eigenvectors to Kronecker product form. Consequently, a small value of $D^{\mathrm{spec}}_{K,\mathrm{kron}}$ is a certificate only for the shared-eigenbasis subproblem unless accompanied by an eigenvector-compatibility argument. For a general Hessian, this missing compatibility can dominate the spectral width calculation: the additive log-spectrum may be favorable while no nearby Kronecker eigenbasis aligns with the curvature directions.

*Double-centering residual..* For $Y \in \mathbb{R}^{m\times n}$, write

$$\bar{Y} = \frac{1}{mn}\sum_{i,j} Y_{ij}, \qquad r_i = \frac{1}{n}\sum_j Y_{ij} - \bar{Y}, \qquad c_j = \frac{1}{m}\sum_i Y_{ij} - \bar{Y},$$

and define the residual

$$E_{ij} = Y_{ij} - \bar{Y} - r_i - c_j.$$

Then $E$ is orthogonal to $\mathcal{A}_{\mathrm{kron}}$ in Frobenius inner product. The quantity

$$\Delta_{\mathrm{kron}}(Y) = \|E\|_F$$

is invariant under Kronecker spectral updates $Y \mapsto Y + Z$ with $Z \in \mathcal{A}_{\mathrm{kron}}$. It is therefore a structural mismatch signal. Turning it into a condition-number lower bound requires additional width or projection assumptions; the residual alone records nonadditivity.

**5.2. Intrinsic Kronecker quotient geometry.** The factorization has a gauge:

$$B \otimes A = (c^{-1}B) \otimes (cA), \qquad c > 0.$$

Thus the intrinsic factor space is a quotient of $\mathbb{S}^n_{++} \times \mathbb{S}^m_{++}$. For an absolutely continuous factor path define

$$X_t = A_t^{-1/2}\dot{A}_t A_t^{-1/2}, \qquad Y_t = B_t^{-1/2}\dot{B}_t B_t^{-1/2}.$$

PROPOSITION 5.4 (Restricted affine-invariant line element). *For $G_t = B_t \otimes A_t$,*

$$\left\|\dot{G}_t\right\|_{G_t,\mathrm{AI}}^2 = n\,\|X_t\|_F^2 + m\,\|Y_t\|_F^2 + 2\operatorname{tr}(X_t)\operatorname{tr}(Y_t).$$

*The direction $X_t = \alpha I_m,\ Y_t = -\alpha I_n$ has zero length and is exactly the factor-scale gauge direction.*

PROPOSITION 5.5 (Distance between Kronecker metrics). *For $G_0 = B_0 \otimes A_0$ and $G_1 = B_1 \otimes A_1$, set*

$$L_A = \log(A_0^{-1/2} A_1 A_0^{-1/2}), \qquad L_B = \log(B_0^{-1/2} B_1 B_0^{-1/2}).$$

*Then the ambient affine-invariant distance is*

$$d_{\mathrm{AI}}(G_0, G_1)^2 = n\,\|L_A\|_F^2 + m\,\|L_B\|_F^2 + 2\operatorname{tr}(L_A)\operatorname{tr}(L_B).$$

The line element identifies the degenerate gauge direction and the length assigned to a specified factor path. The product-distance formula gives exact distances between two Kronecker metrics. Projecting an arbitrary target Hessian onto the Kronecker family is a separate problem handled after the fixed-basis surrogate below.

PROPOSITION 5.6 (Fixed-basis exactness of the spectral surrogate). *Assume $H$, $G_0$, and all feasible $G_t = B_t \otimes A_t$ share a fixed Kronecker eigenbasis. Let $\alpha_i(t) = \log\lambda_i(A_t)$ and $\beta_j(t) = \log\lambda_j(B_t)$, and set $Z_{ij}(t) = \alpha_i(t) + \beta_j(t)$. Then*

$$\left\|\dot{G}_t\right\|_{G_t,\mathrm{AI}}^2 = \left\|\dot{Z}_t\right\|_F^2.$$

*Consequently, in the fixed-basis log-spectrum subproblem, $D_{K,\mathrm{kron}}^{\mathrm{spec}}$ is the intrinsic Kronecker complexity.*

**5.3. Affine-invariant Kronecker projection.** The fixed-basis surrogate can be strengthened without assuming shared eigenvectors. Define

$$\mathcal{K}_{m,n} = \{B \otimes A : A \in \mathbb{S}_{++}^m,\ B \in \mathbb{S}_{++}^n\} \subset \mathbb{S}_{++}^{mn},$$

and the log-Kronecker subspace

$$\mathcal{L}_{m,n} = \{I_n \otimes X + Y \otimes I_m : X \in \mathbb{S}^m,\ Y \in \mathbb{S}^n\}.$$

For $L \in \mathbb{S}^{mn}$, write $L = [L_{pq}]_{p,q=1}^n$ in $m \times m$ blocks and define partial traces

$$\operatorname{Tr}_B(L) = \sum_{p=1}^n L_{pp}, \qquad \operatorname{Tr}_A(L) = [\operatorname{tr}(L_{pq})]_{p,q=1}^n.$$

THEOREM 5.7 (Log-Kronecker mismatch and closed-form projection). *For $M \in \mathbb{S}_{++}^{mn}$,*

$$M \in \mathcal{K}_{m,n} \iff \log M \in \mathcal{L}_{m,n}.$$

*Moreover, the Frobenius projection of $L \in \mathbb{S}^{mn}$ onto $\mathcal{L}_{m,n}$ is*

$$\Pi_{\mathcal{L}} L = \tau I_{mn} + I_n \otimes X_0 + Y_0 \otimes I_m,$$

*where*

$$\tau = \frac{\operatorname{tr} L}{mn}, \qquad X_0 = \frac{1}{n}\operatorname{Tr}_B(L) - \tau I_m, \qquad Y_0 = \frac{1}{m}\operatorname{Tr}_A(L) - \tau I_n.$$

*Thus*

$$\delta^{\log}_{\mathrm{kron}}(M) = \|\log M - \Pi_{\mathcal{L}} \log M\|_F$$

*is a closed-form, eigenvector-aware Kronecker mismatch certificate. It vanishes exactly on* $\mathcal{K}_{m,n}$.

The double-centering residual above is the special case of Theorem 5.7 in a fixed Kronecker eigenbasis. Outside that regime, $\delta^{\log}_{\mathrm{kron}}$ also detects eigenvector incompatibility.

LEMMA 5.8 (Hadamard projection and line-search facts). *Let $M$ be a finite-dimensional Hadamard manifold and let $C \subset M$ be closed and geodesically convex. Every $x \in M$ has a unique metric projection $P_C(x)$. For fixed $x$, the function*

$$f(y) = \frac{1}{2} d(y,x)^2$$

*is 1-strongly geodesically convex on $C$. If the restricted gradient of $f$ is Lipschitz on a compact geodesic neighborhood of an initial sublevel set, Armijo backtracking along the negative restricted gradient terminates at every step and accepts step sizes bounded below on that sublevel set.*

THEOREM 5.9 (Affine-invariant Kronecker projection). *The family $\mathcal{K}_{m,n}$ is a closed totally geodesic submanifold of the SPD cone under $d_{\mathrm{AI}}$. Hence every $M \in \mathbb{S}^{mn}_{++}$ has a unique projection*

$$G_\star = P_{\mathcal{K}}(M) = \operatorname*{arg\,min}_{G \in \mathcal{K}_{m,n}} d_{\mathrm{AI}}(M, G).$$

*If*

$$R_\star = \log(G_\star^{-1/2} M G_\star^{-1/2}),$$

*then*

$$\mathrm{Tr}_B(R_\star) = 0, \qquad \mathrm{Tr}_A(R_\star) = 0.$$

*Conversely, any $G \in \mathcal{K}_{m,n}$ satisfying these two equations equals $G_\star$. Therefore*

$$d_{\mathrm{AI}}(M, \mathcal{K}_{m,n}) = \|R_\star\|_F.$$

Theorem 5.9 is the intrinsic replacement for the fixed-basis surrogate. It gives an implicit but global distance computation. The log-Euclidean certificate remains useful because the exponential-metric-increasing inequality gives

$$d_{\mathrm{AI}}(M, \mathcal{K}_{m,n}) \geq \delta^{\log}_{\mathrm{kron}}(M).$$

**5.4. When nearest geometry is condition-optimal.** The unique AIRM projection need not minimize the Hessian-relative condition number. The following theorem gives the exact compatibility condition and the regimes currently known to force or violate it.

THEOREM 5.10 (Nearest Kronecker geometry versus best preconditioner). *Let*

$$P = P_{\mathcal{K}}(H), \qquad S = P^{-1/2} H P^{-1/2},$$

*and let $E_+$ and $E_-$ be the largest- and smallest-eigenvalue eigenspaces of $S$. Then $P$ belongs to*

$$\operatorname*{arg\,min}_{G \in \mathcal{K}_{m,n}} \log \kappa(G^{-1} H)$$

*if and only if there exist density matrices* $Q_+, Q_- \succeq 0$ *such that*

$$\operatorname{tr} Q_+ = \operatorname{tr} Q_- = 1, \qquad \operatorname{range}(Q_+) \subseteq E_+, \qquad \operatorname{range}(Q_-) \subseteq E_-,$$

*and*

$$\operatorname{Tr}_B(Q_- - Q_+) = 0, \qquad \operatorname{Tr}_A(Q_- - Q_+) = 0.$$

*For simple extreme eigenvalues, writing* $u_+$ *and* $u_-$ *for unit extreme eigenvectors, this says exactly that the two pure states* $u_+u_+^\top$ *and* $u_-u_-^\top$ *have identical marginal density matrices on both tensor factors.*

*If* $S$ *has exactly two distinct eigenvalues, the condition above always holds: the nearest affine-invariant Kronecker geometry is automatically a globally best Kronecker preconditioner. This universal guarantee does not extend to unrestricted richer spectra. In particular, let*

$$A = \operatorname{diag}(1, -1), \qquad B = \operatorname{diag}(2, -\tfrac{1}{2}, -\tfrac{3}{2}), \qquad H = \exp(B \otimes A) \in \mathbb{S}^6_{++}.$$

*Then*

$$P_{\mathcal{K}_{2,3}}(H) = I_6, \qquad \log \kappa(H) = 4,$$

*whereas the Kronecker metric*

$$G = I_3 \otimes \exp(\tfrac{1}{4}A)$$

*satisfies*

$$\log \kappa(G^{-1}H) = \frac{7}{2}.$$

*Thus the unique nearest structured geometry need not be a best structured preconditioner.*

Corollary 5.11 (Two-by-two completeness and minimal separation dimension). *For* $m = n = 2$ *and every* $H \in \mathbb{S}^4_{++}$*, the unique AIRM projection* $P_{\mathcal{K}}(H)$ *is a globally best Kronecker preconditioner. Consequently, apart from a trivial one-dimensional factor, strict separation between AIRM projection and condition-number optimality requires ambient dimension at least six. The* $2 \times 3$ *construction in Theorem 5.10 is therefore dimension-minimal among real two-factor SPD Kronecker families.*

The mechanism is specific to real two-level factors: zero partial traces put the log residual, after local orthogonal conjugation, in a commuting Bell-diagonal form, and every Bell spectral projector has maximally mixed marginals.

Corollary 5.12 (Marginal-mismatch descent certificate). *In the setting of Theorem 5.10, define*

$$\mathcal{D}_\pm = \{Q \succeq 0 : \operatorname{tr} Q = 1,\ \operatorname{range}(Q) \subseteq E_\pm\}$$

*and the compact convex set of tangent subgradients*

$$\mathcal{G}(P) = \{\Pi_{\mathcal{L}}(Q_- - Q_+) : Q_\pm \in \mathcal{D}_\pm\}.$$

*Let* $g_\star$ *be its unique minimum-Frobenius-norm element and set* $\mu_{\mathrm{marg}}(P) = \|g_\star\|_F$*. Then*

$$\mu_{\mathrm{marg}}(P) = 0 \quad \Longleftrightarrow \quad P \in \operatorname*{arg\,min}_{G \in \mathcal{K}_{m,n}} \log \kappa(G^{-1}H).$$

*If* $\mu_{\mathrm{marg}}(P) > 0$*, the Kronecker geodesic*

$$G_t = P^{1/2} \exp(-t g_\star) P^{1/2}$$

*satisfies the strict one-sided descent identity*

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0+} \log \kappa(G_t^{-1}H) = -\mu_{\mathrm{marg}}(P)^2.$$

*For simple extreme eigenvalues,* $g_\star = \Pi_{\mathcal{L}}(u_-u_-^\top - u_+u_+^\top)$. *With multiplicity,* $g_\star$ *is obtained from a finite-dimensional convex conic problem over the two extreme eigenspaces.*

COROLLARY 5.13 (Residual certificate for the Kronecker projection solver). *For* $G \in \mathcal{K}_{m,n}$*, define*

$$R_G(M) = \log(G^{-1/2}MG^{-1/2}), \qquad V(G) = \Pi_{\mathcal{L}} R_G(M).$$

*Then*

$$V(G) = 0 \quad \Longleftrightarrow \quad G = P_{\mathcal{K}}(M),$$

*and*

$$d_{\mathrm{AI}}(G, P_{\mathcal{K}}(M)) \le \|V(G)\|_F\,, \qquad 0 \le f(G) - f(P_{\mathcal{K}}(M)) \le \frac{1}{2}\|V(G)\|_F^2\,,$$

*where* $f(G) = \frac{1}{2}d_{\mathrm{AI}}(G, M)^2$. *Consequently* $\|V(G)\|_F$ *is a certified stopping residual for a projected Riemannian solver.*

COROLLARY 5.14 (Armijo solver for the Kronecker projection). *Fix* $G_0 \in \mathcal{K}_{m,n}$*,* $\bar\eta > 0$*, and* $\gamma, c \in (0,1)$*. Let* $f(G) = \frac{1}{2}d_{\mathrm{AI}}(G, M)^2$*, let*

$$\mathcal{Q}_0 = \{G \in \mathcal{K}_{m,n} : f(G) \le f(G_0)\},$$

*and let* $L_0$ *be a Lipschitz constant for the restricted gradient of* $f$ *on a compact geodesic neighborhood of* $\mathcal{Q}_0$*; such an* $L_0$ *exists in the finite-dimensional SPD cone. At iteration* $r$*, set*

$$V_r = \Pi_{\mathcal{L}} R_{G_r}(M).$$

*If* $V_r = 0$*, stop. Otherwise choose the largest* $\eta_r \in \{\bar\eta, \bar\eta\gamma, \bar\eta\gamma^2, \ldots\}$ *satisfying*

$$f\Big(G_r^{1/2}\exp(\eta_r V_r)G_r^{1/2}\Big) \le f(G_r) - c\eta_r \|V_r\|_F^2\,.$$

*Then*

$$G_{r+1} = G_r^{1/2}\exp(\eta_r V_r)G_r^{1/2}$$

*is well defined, remains in* $\mathcal{K}_{m,n}$*, and either terminates at* $P_{\mathcal{K}}(M)$ *or converges to* $P_{\mathcal{K}}(M)$*. Moreover, there exists* $\underline{\eta} \in (0, (2c)^{-1}]$*, depending only on* $\mathcal{Q}_0, L_0, \bar\eta, \gamma$*, and* $c$*, such that*

$$f(G_r) - f(P_{\mathcal{K}}(M)) \le (1 - 2c\underline{\eta})^r\big[f(G_0) - f(P_{\mathcal{K}}(M))\big].$$

THEOREM 5.15 (Self-conditioned Kronecker $K$-target bounds). *Let*

$$\mathcal{K}_{m,n,K} = \mathcal{K}_{m,n} \cap \mathcal{C}_K.$$

*Here* $\mathcal{C}_K \subset \mathbb{S}_{++}^{mn}$*, so this target constrains the condition number of the Kronecker metric itself. It is an auxiliary self-conditioning target. For a fixed Hessian* $H$*, the RDGC preconditioning target is instead* $\{G \in \mathcal{K}_{m,n} : \kappa(G^{-1}H) \le K\}$*, equivalently* $\mathcal{S}_K(H) \cap \mathcal{C}_K$ *in relative coordinates. The two targets coincide only under additional compatibility assumptions.*

*The self-conditioned set $\mathcal{K}_{m,n,K}$ is closed and geodesically convex. If $G_\star = P_{\mathcal{K}}(M)$, then*

$$d_{\mathrm{AI}}(M, \mathcal{K}_{m,n,K})^2 \geq d_{\mathrm{AI}}(M, \mathcal{K}_{m,n})^2 + d_{\mathrm{AI}}(G_\star, \mathcal{K}_{m,n,K})^2,$$

*and*

$$d_{\mathrm{AI}}(M, \mathcal{K}_{m,n,K}) \leq d_{\mathrm{AI}}(M, \mathcal{K}_{m,n}) + d_{\mathrm{AI}}(G_\star, \mathcal{K}_{m,n,K}).$$

*Moreover, if $G_0 = B_0 \otimes A_0 \in \mathcal{K}_{m,n}$ and*

$$A_0 = U \operatorname{diag}(e^{a_1}, \dots, e^{a_m}) U^\top, \qquad B_0 = V \operatorname{diag}(e^{b_1}, \dots, e^{b_n}) V^\top$$

*with sorted $a_i$ and $b_j$, then*

$$d_{\mathrm{AI}}(G_0, \mathcal{K}_{m,n,K})^2 = \min_{x,y} \sum_{i=1}^{m} \sum_{j=1}^{n} (x_i + y_j - a_i - b_j)^2$$

*subject to*

$$x_1 \leq \cdots \leq x_m, \qquad y_1 \leq \cdots \leq y_n, \qquad (x_m - x_1) + (y_n - y_1) \leq \log K.$$

Theorem 5.16 (Hessian-relative Kronecker reachability). *For a fixed $H \in \mathbb{S}^{mn}_{++}$, define*

$$\mathcal{R}_{K,\mathcal{K}}(H) = \{G \in \mathcal{K}_{m,n} : \kappa(G^{-1}H) \leq K\}.$$

*Then the following are equivalent:*
1. *$\mathcal{R}_{K,\mathcal{K}}(H) \neq \emptyset$;*
2. *$\mathcal{C}_{K,\mathcal{K}}(H) = \mathcal{S}_{\mathcal{K}}(H) \cap \mathcal{C}_K \neq \emptyset$;*
3. *there exist $A \succ 0$, $B \succ 0$, and $\lambda > 0$ such that*

$$\lambda(B \otimes A) \preceq H \preceq K\lambda(B \otimes A).$$

*Equivalently, because the scale $\lambda$ can be absorbed into one factor, reachability is the feasibility of*

$$B \otimes A \preceq H \preceq K(B \otimes A), \qquad A \succ 0, \quad B \succ 0.$$

*If these equivalent conditions hold, fix $G_0 \in \mathcal{K}_{m,n}$ and set $S_0 = H^{-1/2} G_0 H^{-1/2}$. Then*

$$D_{K,\mathcal{K}}(S_0; H) = d_{\mathrm{AI}}\big(G_0, \mathcal{R}_{K,\mathcal{K}}(H)\big).$$

*Moreover, $\mathcal{R}_{K,\mathcal{K}}(H)$ is a closed geodesically convex subset of the Kronecker manifold, and the displayed distance is attained at a unique projection point. Thus the endpoint reachability problem is exact, but the Kronecker Loewner sandwich is generally a nonconvex feasibility problem in the ambient matrix variable.*

*More strongly, suppose a strict feasible certificate $G_s \in \mathcal{K}_{m,n}$ satisfies*

$$\log \kappa(G_s^{-1} H) \leq \log K - \sigma, \qquad \sigma > 0,$$

*and let $R = d_{\mathrm{AI}}(G_0, G_s) > 0$. On $\mathcal{D} = \mathcal{K}_{m,n} \cap \overline{B}_R(G_0)$, every $\Lambda > 2R^2/\sigma$ makes*

$$\frac{1}{2} d_{\mathrm{AI}}(G_0, G)^2 + \Lambda \left[\log \kappa(G^{-1}H) - \log K\right]_+$$

*an exact penalty for the unique target projection. The proximal iteration of Theorem 3.4 converges to that projection with the global squared-distance contraction factor $(1+2\eta)^{-1}$ per step.*

COROLLARY 5.17 (Kronecker expression threshold). *Define*

$$K_{\mathcal{K}}^{\star}(H) = \min_{G \in \mathcal{K}_{m,n}} \kappa(G^{-1}H).$$

*The minimum is attained. Moreover, for every* $K \geq 1$,

$$\mathcal{R}_{K,\mathcal{K}}(H) \neq \emptyset \quad \Longleftrightarrow \quad K \geq K_{\mathcal{K}}^{\star}(H).$$

*Thus* $K_{\mathcal{K}}^{\star}(H)$ *is the intrinsic endpoint threshold for whether a Kronecker metric can reach the Hessian-relative condition-number target.*

THEOREM 5.18 (Residual-calibrated Kronecker threshold bracket). *Assume* $mn \geq 2$. *Let* $H \in \mathbb{S}_{++}^{mn}$, $G \in \mathcal{K}_{m,n}$, *and*

$$P = P_{\mathcal{K}}(H), \qquad R_G(H) = \log(G^{-1/2} H G^{-1/2}), \qquad V(G) = \Pi_{\mathcal{L}} R_G(H).$$

*Set*

$$\eta = \|V(G)\|_F\,, \qquad \rho = d_{\mathrm{AI}}(H, G), \qquad \delta_- = \sqrt{\max\{\rho^2 - \eta^2, 0\}},$$

*and*

$$\kappa_G = \kappa(G^{-1}H), \qquad \alpha_{mn} = \sqrt{\frac{mn}{\lfloor (mn)^2/4 \rfloor}}.$$

*Then*

$$\delta_- \leq d_{\mathrm{AI}}(H, \mathcal{K}_{m,n}) \leq \rho$$

*and*

$$\exp(\alpha_{mn}\delta_-) \leq K_{\mathcal{K}}^{\star}(H) \leq \kappa_G.$$

*Consequently, if* $\kappa_G \leq K$, *then* $G$ *is a primal Hessian-relative Kronecker certificate and*

$$D_{K,\mathcal{K}}(S_0; H) \leq d_{\mathrm{AI}}(G_0, G)$$

*for every* $G_0 \in \mathcal{K}_{m,n}$, *where* $S_0 = H^{-1/2} G_0 H^{-1/2}$. *Moreover, with* $\kappa_P = \kappa(P^{-1}H)$,

$$e^{-2\eta}\kappa_G \leq \kappa_P \leq e^{2\eta}\kappa_G.$$

*Writing* $\delta_\star = d_{\mathrm{AI}}(H, \mathcal{K}_{m,n})$, *the condition-number loss of using the nearest geometry rather than a best preconditioner obeys*

$$0 \leq \log \frac{\kappa_P}{K_{\mathcal{K}}^{\star}(H)} \leq \log \kappa_P - \alpha_{mn}\delta_\star \leq \log \kappa_G + 2\eta - \alpha_{mn}\delta_-.$$

*Thus* $e^{2\eta}\kappa_G \leq K$ *certifies that the exact projection* $P_{\mathcal{K}}(H)$ *reaches the target, while* $e^{-2\eta}\kappa_G > K$ *certifies that this particular projection candidate does not. Finally, if*

$$\exp(\alpha_{mn}\delta_-) > K,$$

*then the full noncommuting Kronecker target* $\mathcal{R}_{K,\mathcal{K}}(H)$ *is empty.*

PROPOSITION 5.19 (Fixed-basis Hessian-relative Kronecker exactness). *Fix orthogonal matrices* $U \in \mathbb{R}^{m \times m}$ *and* $V \in \mathbb{R}^{n \times n}$, *and consider the fixed-basis subfamily*

$$\mathcal{K}_{U,V} = \{(V \operatorname{diag}(e^{b_1}, \ldots, e^{b_n}) V^\top) \otimes (U \operatorname{diag}(e^{a_1}, \ldots, e^{a_m}) U^\top) : a \in \mathbb{R}^m,\ b \in \mathbb{R}^n\}.$$

*Assume*

$$H = (V \otimes U) \operatorname{diag}(h_{ij}) (V \otimes U)^\top, \qquad h_{ij} > 0,$$

*and set* $\ell_{ij} = \log h_{ij}$*. Then reachability in* $\mathcal{K}_{U,V}$ *at target* $K$ *is equivalent to the linear feasibility problem*

$$\exists a \in \mathbb{R}^m,\ b \in \mathbb{R}^n,\ c \in \mathbb{R} \quad \textit{such that} \quad c \le \ell_{ij} - a_i - b_j \le c + \log K \quad \forall i, j.$$

*Equivalently, the fixed-basis threshold is*

$$\log K^\star_{U,V}(H) = \min_{a,b} \left[ \max_{i,j}(\ell_{ij} - a_i - b_j) - \min_{i,j}(\ell_{ij} - a_i - b_j) \right].$$

*Feasibility for this fixed-basis subfamily is a primal certificate for the full Kronecker family. Infeasibility for the fixed-basis subfamily does not rule out a noncommuting Kronecker metric with different factor eigenvectors.*

Theorem 5.20 (Fixed-basis primal–dual Kronecker obstruction). *In the setting of Proposition 5.19, assume* $m, n \ge 2$ *and let* $L = (\ell_{ij})$*. The optimal fixed-basis log-threshold*

$$\tau^\star_{U,V}(H) = \min_{a,b} \left[ \max_{i,j}(\ell_{ij} - a_i - b_j) - \min_{i,j}(\ell_{ij} - a_i - b_j) \right]$$

*has the exact dual representation*

$$\tau^\star_{U,V}(H) = \max_{W} \langle W, L \rangle,$$

*where the maximum is over* $W \in \mathbb{R}^{m \times n}$ *satisfying*

$$\sum_j W_{ij} = 0 \quad \forall i, \qquad \sum_i W_{ij} = 0 \quad \forall j, \qquad \sum_{i,j} |W_{ij}| \le 2.$$

*Equivalently, any nonzero optimal witness can be normalized so that* $\sum_{ij} W^+_{ij} = \sum_{ij} W^-_{ij} = 1$*. Hence any feasible dual witness with*

$$\langle W, L \rangle > \log K$$

*certifies that the fixed-basis Kronecker target is infeasible.*

Theorem 5.21 (Soundness of KRON-CERTIFY). *Consider the following certificate procedure. If a fixed Kronecker eigenbasis is supplied, solve the primal–dual fixed-basis LP in Proposition 5.19 and Theorem 5.20. It records the fixed-basis feasible status for a primal solution, and the fixed-basis obstruction status when a feasible dual witness satisfies* $\langle W, L \rangle > \log K$*.*

*In the full noncommuting branch, compute* $\rho, \eta, \delta_-$*, and* $\kappa_G$ *for an iterate* $G \in \mathcal{K}_{m,n}$ *as in Theorem 5.18. The four non-inconclusive tests are*

1. GLOBAL_INFEASIBLE *if* $\exp(\alpha_{mn}\delta_-) > K$*;*
2. FEASIBLE_DIRECT *if* $\kappa_G \le K$*;*
3. PROJECTED_FEASIBLE *if* $e^{2\eta}\kappa_G \le K$*;*
4. PROJECTED_INFEASIBLE *if* $e^{-2\eta}\kappa_G > K$*.*

*If no certificate fires, the procedure records* INCONCLUSIVE*.*

*Every non-inconclusive statement returned by this procedure is correct, with the stated scope: full-family infeasibility, full-family direct feasibility, exact-projection feasibility or infeasibility, fixed-basis feasibility, or fixed-basis obstruction.*

COROLLARY 5.22 (Interval-safe full Kronecker certification). *Let the exact quantities in Theorem 5.18 have validated enclosures*

$$0 \le \underline{\rho} \le \rho, \qquad 0 \le \eta \le \overline{\eta}, \qquad 1 \le \underline{\kappa} \le \kappa_G \le \overline{\kappa}.$$

*Set*

$$\underline{\delta} = \sqrt{\max\{\underline{\rho}^2 - \overline{\eta}^2, 0\}}.$$

*If the displayed exponentials, products, and final comparisons are also evaluated with outward rounding, equivalently as validated log-domain comparisons, then each of the following finite-precision tests is sound:*

$$\begin{aligned} \exp(\alpha_{mn}\underline{\delta}) > K &\implies \mathcal{R}_{K,\mathcal{K}}(H) = \emptyset, \\ \overline{\kappa} \le K &\implies G \in \mathcal{R}_{K,\mathcal{K}}(H), \\ e^{2\overline{\eta}}\overline{\kappa} \le K &\implies P_{\mathcal{K}}(H) \in \mathcal{R}_{K,\mathcal{K}}(H), \\ e^{-2\overline{\eta}}\underline{\kappa} > K &\implies P_{\mathcal{K}}(H) \notin \mathcal{R}_{K,\mathcal{K}}(H). \end{aligned}$$

*If none fires, the interval-safe full-family branch returns* INCONCLUSIVE. *The enclosures may come from interval arithmetic or from certified backward-error bounds for the eigensolver and matrix logarithm; unvalidated floating-point estimates alone do not meet the hypothesis.*

PROPOSITION 5.23 (Hessian-relative Kronecker candidate certificate). *Let $H \in \mathbb{S}^{mn}_{++}$, $G_0 = B_0 \otimes A_0 \in \mathcal{K}_{m,n}$, and $G_c = B_c \otimes A_c \in \mathcal{K}_{m,n}$. Define*

$$S_0 = H^{-1/2} G_0 H^{-1/2}, \qquad S_c = H^{-1/2} G_c H^{-1/2}.$$

*If*

$$\kappa(S_c) \le K,$$

*equivalently $\kappa(G_c^{-1}H) \le K$, then $G_c$ is a primal Hessian-relative Kronecker certificate and*

$$D_{K,\mathcal{K}}(S_0; H) \le d_{\mathrm{AI}}(G_0, G_c).$$

*With*

$$L_A = \log(A_0^{-1/2} A_c A_0^{-1/2}), \qquad L_B = \log(B_0^{-1/2} B_c B_0^{-1/2}),$$

*this upper bound has the closed form*

$$d_{\mathrm{AI}}(G_0, G_c)^2 = n\,\|L_A\|_F^2 + m\,\|L_B\|_F^2 + 2\operatorname{tr}(L_A)\operatorname{tr}(L_B).$$

*Thus any candidate Kronecker metric passing the generalized condition-number test supplies a finite restricted-complexity upper bound. If a proposed candidate fails the test, no infeasibility conclusion follows.*

Taking $G_c = P_{\mathcal{K}}(H)$ gives a projection-based sufficient certificate: one checks the Hessian-relative condition number of the projected Kronecker metric and, if it is at most $K$, obtains the path-length upper bound above. Failure of this projected candidate only means that this particular candidate does not reach the RDGC target; another Kronecker metric may still do so.

PROPOSITION 5.24 (Nonsmooth active $K$-target condition). *Assume $K > 1$. Let $Q_\star = P_{\mathcal{K}_{m,n,K}}(M)$, and set*

$$R_\star = \log(Q_\star^{-1/2} M Q_\star^{-1/2}), \qquad \omega(Q) = \log\kappa(Q).$$

*If* $\omega(Q_\star) < \log K$, *then*

$$\Pi_{\mathcal{L}} R_\star = 0.$$

*If* $\omega(Q_\star) = \log K$, *then there exist* $\mu \geq 0$ *and* $W_\star \in \partial\omega(Q_\star)$ *such that*

$$\Pi_{\mathcal{L}} R_\star = \mu \Pi_{\mathcal{L}} W_\star.$$

*Here* $\partial\omega(Q_\star)$ *denotes the subdifferential in the whitened AIRM tangent coordinate* $X \mapsto \omega(Q_\star^{1/2} \exp(X) Q_\star^{1/2})$ *at* $X = 0$. *It is the convex hull of*

$$uu^\top - vv^\top, \qquad u \in E_{\max},\ \|u\| = 1, \quad v \in E_{\min},\ \|v\| = 1,$$

*with* $E_{\max}$ *and* $E_{\min}$ *the extremal eigenspaces of* $Q_\star$. *For simple extremal eigenvalues this reduces to the usual multiplier equation with* $W_\star = u_{\max} u_{\max}^\top - u_{\min} u_{\min}^\top$.

PROPOSITION 5.25 (Projection diagnostics for K-FAC-style metrics). *Suppose an optimizer uses damped Kronecker metric states*

$$G_t = (B_t + \lambda_B I_n) \otimes (A_t + \lambda_A I_m).$$

*Let* $H_t \succ 0$ *be a reference curvature and* $P_t = P_{\mathcal{K}}(H_t)$. *Then the quantities*

$$M_t = d_{\mathrm{AI}}(H_t, P_t), \quad A_t = d_{\mathrm{AI}}(P_t, G_t), \quad E_t = d_{\mathrm{AI}}(H_t, G_t), \quad C_{K,t} = d_{\mathrm{AI}}(P_t, \mathcal{K}_{m,n,K})$$

*satisfy*

$$E_t^2 \geq M_t^2 + A_t^2,$$

*and*

$$(M_t^2 + C_{K,t}^2)^{1/2} \leq d_{\mathrm{AI}}(H_t, \mathcal{K}_{m,n,K}) \leq M_t + C_{K,t}.$$

*Here* $A_t$ *and the discrete increment* $d_{\mathrm{AI}}(G_t, G_{t+1})$ *have the closed form of Proposition* 5.5, $M_t$ *is computed by Theorem* 5.9, *and* $C_{K,t}$ *is the convex QP in Theorem* 5.15. *These quantities diagnose distance to the Kronecker family, optimizer-state lag inside that family, and distance to the auxiliary self-conditioned target. They do not by themselves certify* $\kappa(G_t^{-1} H_t) \leq K$. *A full preconditioning certificate must use the Hessian-relative target* $\{G \in \mathcal{K}_{m,n} : \kappa(G^{-1} H_t) \leq K\}$ *and must specify how the optimizer state, damping, bias correction, and discrete path realize an admissible family trajectory.*

**5.5. Scoped low-rank extension.** Low-rank structures split into three different models:

1. additive metric families, $G = D + UU^\top$, which are close to practical low-rank preconditioners;
2. log-low-rank families, $G = \exp(cI + L)$ with $\mathrm{rank}(L) \leq r$, which are cleaner for spectral theory;
3. inverse Woodbury forms, useful for implementation cost.

This paper uses low-rank as an extension layer. The result below is a spectral surrogate for practical additive forms $D + UU^\top$; their intrinsic geometry requires a separate stratified analysis. The clean spectral model assumes a shared eigenbasis and lets at most $r$ log-eigenvalue coordinates move away from an overall shift:

$$\mathcal{Y}_{K,r}(y) = \{z \in \mathcal{Y}_K : \exists c \in \mathbb{R},\ |\{i : z_i \neq y_i + c\}| \leq r\}.$$

The corresponding spectral low-rank complexity is

$$D_{K,r}^{\mathrm{spec}}(y) = \mathrm{dist}_{\ell^2}(y, \mathcal{Y}_{K,r}(y)).$$

PROPOSITION 5.26 (Low-rank spectral monotonicity). *If $r_1 \le r_2$, then*

$$D^{\mathrm{spec}}_{K,r_1}(y) \ge D^{\mathrm{spec}}_{K,r_2}(y) \ge D_K(y).$$

*When $r \ge d$, the low-rank spectral complexity equals the full spectral benchmark $D_K(y)$.*

*Remark* 5.27 (Role of low rank). The proposition captures the intended hierarchy: more spectral correction directions cannot hurt, and full rank recovers the unconstrained benchmark. For practical additive forms $D + UU^\top$, the same monotonicity should be studied in a stratified metric space rather than in this shared-eigenbasis spectral surrogate. The proposition therefore supports the hierarchy of spectral correction families under the stated shared-eigenbasis surrogate. Claims about a concrete low-rank preconditioner additionally require its update parameterization, damping convention, and admissible interpolation to be mapped into one of these geometric realizations.

## 6. Robustness Interfaces and Certificate Verification.

### 6.1. Loewner proxy robustness.

PROPOSITION 6.1 (Proxy inflation). *If*

$$(1-\varepsilon)H \preceq \widehat{H} \preceq (1+\varepsilon)H, \qquad 0 \le \varepsilon < 1,$$

*then every $G \succ 0$ satisfies*

$$\kappa(G^{-1}H) \le \frac{1+\varepsilon}{1-\varepsilon}\kappa(G^{-1}\widehat{H}).$$

*Consequently, reaching the proxy threshold*

$$\widehat{K} = K\frac{1-\varepsilon}{1+\varepsilon}$$

*is sufficient for the true threshold whenever $\widehat{K} \ge 1$.*

PROPOSITION 6.2 (High-probability proxy interface). *Suppose a random proxy obeys the displayed Loewner event with probability at least $1-\delta$. If a structured candidate $G$ satisfies $\kappa(G^{-1}\widehat{H}) \le \widehat{K}$ on that event, then $\kappa(G^{-1}H) \le K$ with probability at least $1-\delta$.*

The second statement is an interface. A finite-sample result requires a concentration theorem for the chosen estimator [40].

**6.2. Deterministic and randomized checks.** The accompanying code supplement contains deterministic and randomized small-SPD checks for diagonal and block certificates, Kronecker spectral mismatch, Hessian-relative candidates, projection residuals, and strict separation. The entry points are

- `synthetic_certificate_benchmark.py`,
- `random_spd_certificate_suite.py`,
- `kron_certify_demo.py`.

Failed identities raise assertions. The reference environment is Python 3.11.4 with NumPy 1.26.4, SciPy 1.10.1, Matplotlib 3.7.1, and SymPy 1.11.1.

In the randomized suite, seeds $0, \dots, 63$ are used. The diagonal target $K = 3$ is reachable in 32 instances, the block-Jacobi target $K = 3$ is certified in 58, and the fixed-basis Kronecker spectral target $K = 3$ is reachable in 57. All 64 constructed

Hessian-relative Kronecker positive controls reach $K = 2$, while the identity constructed negative control fails that target in all 64 cases.

The positive controls use

$$H = G^{1/2} C G^{1/2}, \qquad \kappa(C) \in [1.1, 1.8],$$

with target $K = 2$. The counts therefore measure theorem-check coverage in the declared suite, not population success probabilities.

For an independent projection check, the noncommuting $2 \times 2$ example is also optimized by generic BFGS in a five-dimensional gauge-fixed log-factor chart. Three deterministic starts agree on the objective to less than $2 \times 10^{-15}$. The partial-trace Armijo iterate stops at residual $8.257 \times 10^{-6}$; its objective exceeds the independent value by $3.227 \times 10^{-11}$, below the theorem's residual bound $3.409 \times 10^{-11}$, and the two matrices are AIRM distance $7.823 \times 10^{-6}$ apart. BFGS uses gradient tolerance $10^{-7}$; Armijo uses initial step one, shrink factor $1/2$, parameter $10^{-4}$, and residual tolerance $10^{-4}$. These calculations test the theorem through a distinct parameterization and optimizer; they do not establish a large-scale complexity claim.

The same deterministic script directly checks the marginal criterion in Theorem 5.10 and Corollaries 5.11 and 5.12. For the exact $2 \times 3$ separation, the projection normal residual is zero and $\mu_{\mathrm{marg}} = 0.816496580927726$. Hence the predicted one-sided descent slope is $-\mu_{\mathrm{marg}}^2 = -2/3$; a step of $10^{-6}$ gives the finite-difference slope $-0.666666666759852$. In a deterministic $2 \times 2$ Bell-canonical instance with projection log-condition 2.2, the projection normal residual is zero and the computed marginal mismatch is $3.21 \times 10^{-16}$. These two checks exercise respectively the strict separation and the low-dimensional completeness mechanisms.

The deterministic sensitivity scan varies the diagonal coupling and target $K$, block coupling, fixed-basis Kronecker threshold, and low-rank spectral budget. It contains 86 finite and 49 unreachable diagonal grid entries; the maximum block primal and dual numerical violations are $2.21 \times 10^{-16}$ and zero. The low-rank reachable threshold decreases monotonically from 121.51 at rank zero to 1 at full rank, as required by Proposition 5.26. These parameter scans expose certificate branch changes and numerical residuals rather than estimating a data-generating success probability.

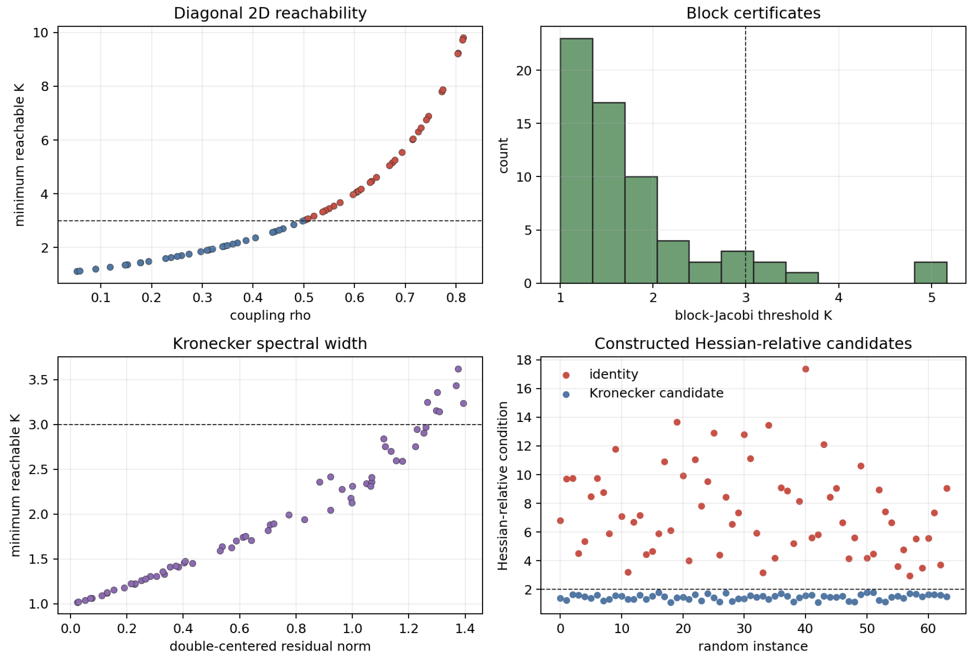


FIG. 1. *Randomized small-SPD certificate checks over 64 seeds. The panels record diagonal reachability, block-Jacobi thresholds, fixed-basis Kronecker spectral width, and constructed Hessian-relative Kronecker candidates. The identity is a constructed negative control. These are theorem checks, not optimizer comparisons.*

**7. Discussion.** The matrix results separate expressivity, approximation, and preconditioning quality. LMI feasibility determines whether a diagonal or block family can reach a condition target. Affine-invariant projection determines the nearest structured geometry. Condition-number minimization selects the best preconditioner. These operations agree in special regimes and separate in general.

For Kronecker matrices, the factor scale gauge is part of the geometry rather than a numerical nuisance. Quotienting the gauge gives a complete totally geodesic manifold, a closed-form intrinsic line element, and a unique AIRM projection. The partial-trace normal equations expose the projection residual, while the extreme-state marginal condition determines whether that projection is also condition-optimal. The marginal-mismatch residual gives a computable first-order measure of failure. Every two-level relative spectrum forces projection optimality. Every $2 \times 2$ Kronecker projection is also condition-optimal, making the explicit $2 \times 3$ separation dimension-minimal. Thus the universal guarantee does not extend to unrestricted richer spectra in the next possible factor dimension.

At the tangent-space level, the same orthogonality calculation extends to a product of $k$ SPD factors. In whitened coordinates its tangent log space is the sum of the $k$ single-mode terms

$$I \otimes \cdots \otimes I \otimes X_j \otimes I \otimes \cdots \otimes I.$$

Repeating the two-factor normal-space proof shows that the projection equations require every one-mode partial trace of the log residual to vanish, and that projection optimality is equivalent to choosing admissible density matrices in the two extreme eigenspaces with matching marginals on every mode. Fixed-basis dual formulas and dimension-minimal separation for three or more factors require separate analysis; the two-factor classification proved here does not presume either.

The exact penalty and proximal contraction provide a global variational baseline; their rate is conditional on solving each proximal subproblem. Likewise, the Armijo theorem gives a convergent projection solver but does not claim that a dense Kronecker projection is inexpensive at neural-network scale. Practical implementations require matrix-free partial traces, structured eigensolvers, and estimator-specific concentration bounds.

The strongest statements are deterministic finite-dimensional matrix theorems. Time-dependent curvature, damping, and optimizer states can be inserted only after a concrete algorithm-to-geometry realization is specified. The Loewner proxy results state precisely what must be supplied by such a statistical layer.

The analytic certificates are exact-arithmetic statements. The interval-safe form remains rigorous when distance, residual, and condition number are enclosed by validated numerical bounds and the derived comparisons are evaluated with outward rounding. Ordinary floating-point residuals without such enclosures remain diagnostics; near multiple extreme eigenvalues, the marginal condition should be solved over the full extreme eigenspaces rather than from a single numerically selected eigenvector.

In summary, structured preconditioning has three distinct certificates: reachability of a target, proximity to a full geometry, and condition-optimality inside the family. Affine-invariant matrix geometry places all three in one framework and makes their coincidence and separation mathematically explicit.

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## Appendix A. Proofs for Spectral Targets and Structured Families.

*Proof of Theorem 3.1.* Let $S_1 \in \mathcal{C}_K$, and let $z_1 \le \cdots \le z_d$ be the ordered log-eigenvalues of $S_1$. The affine-invariant distance satisfies the spectral lower bound

$$d_{\mathrm{AI}}(S_0, S_1) \ge \|y - z\|_2 ,$$

where $y$ is the ordered log-spectrum of $S_0$. This is the standard eigenvalue majorization lower bound for the affine-invariant metric on positive-definite matrices [6]. Since $S_1 \in \mathcal{C}_K$, the vector $z$ has width at most $\log K$, so $z_i \in [c, c + \log K]$ for some $c$. Hence

$$d_{\mathrm{AI}}(S_0, S_1)^2 \ge \sum_{i=1}^{d} \mathrm{dist}(y_i, [c, c + \log K])^2.$$

Taking the infimum over $S_1 \in \mathcal{C}_K$ gives the lower bound.

For the reverse inequality, fix $c$ and define

$$z_i(c) = \Pi_{[c,c+\log K]}(y_i),$$

the scalar projection of $y_i$ onto the interval. The vector $z(c)$ has width at most $\log K$. Choose $S_1$ with the same eigenvectors as $S_0$ and log-eigenvalues $z_i(c)$. Then $S_1 \in \mathcal{C}_K$

and the commuting affine-invariant distance is exactly $\|y - z(c)\|_2$. Minimizing over $c \in \mathbb{R}$ proves the formula. □

*Proof of Proposition 3.8.* Under the stated compatibility assumption, every admissible $\mathfrak{F}_1$-path from the common initial point $S_0$ to the $\mathfrak{F}_1$-target is also an admissible $\mathfrak{F}_2$-path with the same length. Moreover $\mathcal{C}_{K,\mathfrak{F}_1}(H) \subseteq \mathcal{C}_{K,\mathfrak{F}_2}(H)$. Thus the feasible path set for $\mathfrak{F}_2$ contains the feasible path set for $\mathfrak{F}_1$, and the infimum cannot increase. Enlarging the target threshold $K$ similarly enlarges the feasible endpoint set. The ambient SPD cone contains every admissible restricted path considered in the lower-bound statement, giving $D_{K,\mathfrak{F}} \geq D_K$. □

*Proof of Proposition 3.9.* Under the smooth embedded submanifold assumption, admissible paths are exactly absolutely continuous paths in $\mathcal{S}_{\mathfrak{F}}(H)$ from $S_0$ to $\mathcal{C}_{K,\mathfrak{F}}(H)$. The induced length is the ambient affine-invariant length restricted to that submanifold. Hence the infimum is the intrinsic distance to the target. If no path connects the component of $S_0$ to the target, the intrinsic distance is $+\infty$. □

*Proof of Corollary 3.7.* We first record the sharp vector inequality used below. For $x \in \mathbb{R}^d$, write

$$\operatorname{osc}(x) = \max_i x_i - \min_i x_i, \qquad \operatorname{dist}(x, \mathbb{R}\mathbf{1}) = \min_{s\in\mathbb{R}} \|x - s\mathbf{1}\|_2 .$$

Let $\tau = \operatorname{osc}(x)$. Translating $x$ by a scalar multiple of $\mathbf{1}$ changes neither side, so assume $\min_i x_i = 0$ and $\max_i x_i = \tau$. Then $0 \leq x_i \leq \tau$. The squared distance to $\mathbb{R}\mathbf{1}$ is $\sum_i (x_i - \bar{x})^2$, which is the variance numerator. Among vectors in the box $[0, \tau]^d$, this convex function is maximized at a vertex. If $k$ coordinates equal $\tau$ and $d - k$ equal 0, then

$$\sum_i (x_i - \bar{x})^2 = \frac{k(d-k)}{d}\tau^2 \leq \frac{\lfloor d^2/4 \rfloor}{d}\tau^2.$$

Thus

$$\operatorname{osc}(x) \geq \sqrt{\frac{d}{\lfloor d^2/4 \rfloor}}\, \operatorname{dist}(x, \mathbb{R}\mathbf{1}).$$

The constant is attained by placing $\lfloor d/2 \rfloor$ coordinates at one endpoint and the remaining coordinates at the other endpoint.

Now fix $Q \in \mathfrak{F}$ and let

$$\lambda_i(Q^{-1/2} H Q^{-1/2}) = \exp(\ell_i).$$

Then

$$\log \kappa(Q^{-1}H) = \operatorname{osc}(\ell).$$

Because $\mathfrak{F}$ is scale closed, $cQ \in \mathfrak{F}$ for every $c > 0$. The eigenvalues of $H^{-1/2}(cQ)H^{-1/2}$ are $\exp(\log c - \ell_i)$, so

$$\inf_{c>0} d_{\mathrm{AI}}(H, cQ) = \operatorname{dist}(\ell, \mathbb{R}\mathbf{1}).$$

The vector inequality gives

$$\log \kappa(Q^{-1}H) \geq \alpha_d \inf_{c>0} d_{\mathrm{AI}}(H, cQ) \geq \alpha_d d_{\mathrm{AI}}(H, \mathfrak{F}).$$

Taking the infimum over $Q \in \mathfrak{F}$ proves the lower bound. Sharpness follows from the scale family $\mathfrak{F} = \{cI : c > 0\}$ and any equality vector in the sharp oscillation-distance inequality, used as the log-spectrum of $H$. □

*Proof of Theorems* 3.3 *and* 3.4. We first prove the log-majorization statement that drives the theorem. For $A, B \in \mathbb{S}^d_{++}$, $t \in [0,1]$, and $1 \le k \le d$, the exterior-power representation commutes with the weighted geometric mean:

$$\bigwedge^k (A\#_t B) = \left(\bigwedge^k A\right) \#_t \left(\bigwedge^k B\right).$$

For positive-definite $X, Y$, monotonicity of the geometric mean gives

$$\|X\#_t Y\|_{\mathrm{op}} \le \|X\|_{\mathrm{op}}^{1-t} \|Y\|_{\mathrm{op}}^{t},$$

because $X \preceq \|X\|_{\mathrm{op}} I$ and $Y \preceq \|Y\|_{\mathrm{op}} I$. Applying this inequality to the exterior powers yields

$$\prod_{i=1}^k \lambda_i^{\downarrow}(A\#_t B) \le \prod_{i=1}^k \lambda_i^{\downarrow}(A)^{1-t} \lambda_i^{\downarrow}(B)^t.$$

Equality holds for $k = d$ because $\det(A\#_t B) = \det(A)^{1-t}\det(B)^t$. Taking logarithms proves

$$\log\lambda(A\#_t B) \prec (1-t)\log\lambda(A) + t\log\lambda(B), \tag{A.1}$$

up to the irrelevant common choice of decreasing order.

Fix an index $a$, abbreviate $H = H_a$ and $\phi = \phi_a$, and set

$$s_H(G) = \log\lambda(G^{-1/2} H G^{-1/2}).$$

Now fix $G_0, G_1 \in \mathfrak{F}$ and let $G_t = G_0 \#_t G_1$. Congruence by $H^{-1/2}$ is an affine-invariant isometry and preserves geometric means, so with $A_i = H^{-1/2} G_i H^{-1/2}$ we have

$$H^{-1/2} G_t H^{-1/2} = A_0 \#_t A_1.$$

The eigenvalues of $G^{-1/2} H G^{-1/2}$ are the reciprocals of those of $H^{-1/2} G H^{-1/2}$. Majorization is preserved by multiplication by $-1$, and a convex permutation-invariant function is Schur convex. Therefore (A.1) and ordinary convexity of $\phi$ give

$$\begin{aligned}\Omega_{\phi,H}(G_t) &\le \phi((1-t)s_H(G_0) + t s_H(G_1))\\ &\le (1-t)\Omega_{\phi,H}(G_0) + t\Omega_{\phi,H}(G_1).\end{aligned}$$

This proves geodesic convexity of each component $\Omega_a$. Since $\rho$ is coordinatewise nondecreasing,

$$\rho(\Omega_1(G_t), \dots, \Omega_q(G_t)) \le \rho((1-t)\Omega(G_0) + t\Omega(G_1)),$$

and convexity of $\rho$ proves geodesic convexity of $\mathcal{J}$. All functions in the statement are finite convex functions on Euclidean spaces and hence continuous, so $\mathcal{J}$ is closed.

If $\phi_a(x + c\mathbf{1}) = \phi_a(x)$, then

$$\log\lambda((cG)^{-1/2} H_a (cG)^{-1/2}) = s_{H_a}(G) - (\log c)\mathbf{1},$$

which proves scale invariance. For stability, the standard affine-invariant spectral inequality gives

$$\left\| s_{H_a}(G) - s_{\widehat{H}_a}(G) \right\|_2 \le d_{\mathrm{AI}}(H_a, \widehat{H}_a),$$

because congruence by $G^{-1/2}$ is an isometry. The Lipschitz bounds for $\phi_a$ and $\rho$, followed by Cauchy–Schwarz, give the displayed uniform perturbation estimate.

We now prove the static projection theorem. For quadratic kinetic action, $U = 0$ and $\Gamma = \iota_{\mathcal{C}_\tau}$, every curve over the remaining horizon $h = T - t$ satisfies

$$\frac{1}{2}\int_t^T \|\dot\gamma\|^2 \,\mathrm{d}s \geq \frac{\mathrm{Len}(\gamma)^2}{2h} \geq \frac{d(G, \mathcal{C}_\tau)^2}{2h}.$$

Equality is attained by the constant-speed geodesic to the metric projection, which proves the closed form for $V$.

A sublevel set of a closed geodesically convex function on a closed geodesically convex family is closed and geodesically convex. In a finite-dimensional Hadamard manifold it has a unique metric projection: existence follows by restricting a minimizing sequence to a closed bounded ball and applying Hopf–Rinow, while uniqueness follows from strict geodesic convexity of squared distance. The geodesic from $G_0$ to the projection stays in $\mathfrak{F}$ and reaches the target with length equal to the metric distance. Every admissible curve has length at least the distance between its endpoints, proving the least-action identity and the uniqueness of the constant-speed minimizer. The displayed Pythagorean inequality is the standard projection inequality in a Hadamard manifold. Family and threshold monotonicity follow from inclusion of the corresponding target and path sets. Finally,

$$|\widehat{\mathcal{J}} - \mathcal{J}| \leq \varepsilon$$

immediately gives the two target inclusions; applying target monotonicity gives the perturbation bracket for the dynamic cost.

It remains to prove exactness and convergence of the algorithm. Let $P = P_\tau$. Since $G_s$ is feasible, $d(G_0, P) \leq R$, so $P \in \mathcal{D}$. Closed balls in a Hadamard manifold are geodesically convex, hence $\mathcal{D}$ is closed and geodesically convex. Write

$$f(G) = \frac{1}{2} d(G_0, G)^2, \qquad h(G) = \mathcal{J}(G) - \tau.$$

Take any infeasible $G \in \mathcal{D}$, set $a = h(G) > 0$, and follow the geodesic from $G$ to $G_s$. At

$$t = \frac{a}{a + \sigma}$$

the point $Y = G \#_t G_s$ is feasible, because geodesic convexity gives $h(Y) \leq (1-t)a - t\sigma = 0$. Both endpoints lie in the radius-$R$ ball, so

$$d(G, Y) = t\, d(G, G_s) \leq \frac{2Ra}{a + \sigma} \leq \frac{2Ra}{\sigma}.$$

On this ball the function $f$ is $R$-Lipschitz, since

$$|f(X) - f(Y)| \leq \frac{d(G_0, X) + d(G_0, Y)}{2} d(X, Y) \leq R d(X, Y).$$

Consequently

$$\Phi_\Lambda(G) = f(G) + \Lambda a \geq f(Y) + \left(\Lambda - \frac{2R^2}{\sigma}\right) a > f(Y) \geq f(P) = \Phi_\Lambda(P).$$

No infeasible point minimizes $\Phi_\Lambda$. On the feasible set the penalty vanishes and $P$ uniquely minimizes $f$, proving exactness.

The function $f$ is 1-strongly geodesically convex on a Hadamard manifold, while $[h]_+ = \max\{h, 0\}$ is geodesically convex. Thus $\Phi_\Lambda$ is 1-strongly geodesically convex on $\mathcal{D}$. Each proximal objective is proper, closed, and $(1+\eta^{-1})$-strongly geodesically convex, so it has a unique minimizer $G_{r+1} \in \mathcal{D}$. Strong convexity at the two minimizers $P = \arg\min_{\mathcal{D}} \Phi_\Lambda$ and $G_{r+1}$ gives

$$\begin{aligned}
&\Phi_\Lambda(P) + \frac{1}{2\eta} d(G_r, P)^2 \\
&\quad \geq \Phi_\Lambda(G_{r+1}) + \frac{1}{2\eta} d(G_r, G_{r+1})^2 + \frac{1+\eta^{-1}}{2} d(G_{r+1}, P)^2 \\
&\quad \geq \Phi_\Lambda(P) + \frac{1}{2} d(G_{r+1}, P)^2 + \frac{1}{2\eta} d(G_r, G_{r+1})^2 + \frac{1+\eta^{-1}}{2} d(G_{r+1}, P)^2.
\end{aligned}$$

After cancellation and multiplication by $2\eta$,

$$d(G_r, P)^2 \geq (1+2\eta) d(G_{r+1}, P)^2 + d(G_r, G_{r+1})^2.$$

Dropping the last term and iterating proves the stated global contraction. ∎

*Proof of Lemma 3.5.* Put $Y = \log S$ and

$$S_t = e^{-tX/2} S e^{-tX/2}, \qquad \dot{S}_0 = -\frac{1}{2}(XS + SX).$$

The matrix logarithm is Fréchet differentiable on the SPD cone, so

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \log S_t = D\log_S[\dot{S}_0].$$

For the finite convex spectral function $F$, convex directional calculus gives

$$F'(Y; D\log_S[\dot{S}_0]) = \max_{Z \in \partial F(Y)} \operatorname{tr}\Big(Z D\log_S[\dot{S}_0]\Big).$$

The spectral subdifferential theorem [24] implies that every $Z \in \partial F(Y)$ commutes with $Y$, hence also with $S = e^Y$. The Fréchet derivative $D\log_S$ is self-adjoint in the Frobenius inner product, and commutation gives

$$D\log_S[Z] = S^{-1}Z.$$

Therefore

$$\begin{aligned}
\operatorname{tr}\Big(Z D\log_S[\dot{S}_0]\Big) &= \operatorname{tr}\Big(S^{-1} Z \dot{S}_0\Big) \\
&= -\frac{1}{2}\operatorname{tr}\big(S^{-1}ZXS + S^{-1}ZSX\big) \\
&= -\operatorname{tr}(ZX).
\end{aligned}$$

Substitution into the support-function formula proves the directional identity. Since a closed convex subdifferential is determined by its support function, the whitened AIRM subdifferential is exactly $-\partial F(Y)$, with no simplicity assumption on the spectrum. ∎

*Proof of Theorem 3.6.* Represent a tangent direction at $G$ in whitened coordinates by $X \in \widehat{T}_G\mathfrak{F}$, so the corresponding geodesic is $G_t = G^{1/2}\exp(tX)G^{1/2}$. The generalized eigenvalues of $(H_a, G_t)$ are the eigenvalues of

$$\exp(-tX/2) S_a \exp(-tX/2).$$

Applying Lemma 3.5 gives, at simple or repeated spectra,

$$\partial\Omega_a(G) = -\mathfrak{Z}_a(G)$$

in whitened AIRM coordinates.

The finite convex composition rule for the coordinatewise nondecreasing aggregator $\rho$ now yields

$$\partial\mathcal{J}(G) = \left\{ -\sum_{a=1}^{q} \theta_a Z_a : \theta \in \partial\rho(\Omega(G)),\ Z_a \in \mathfrak{Z}_a(G) \right\}$$

in whitened coordinates. On the assumed closed geodesically convex smooth submanifold, the geodesic subgradient inequality shows that an ambient subgradient orthogonal to the tangent space is globally sufficient. The constrained Fermat rule gives the converse. Since a normal space is a linear space, the minus sign is immaterial, proving the aggregate normal-response criterion.

For $P = P_{\mathfrak{F}}(H)$, the whitened gradient at $P$ of the squared-distance objective is

$$-\log(P^{-1/2} H P^{-1/2}).$$

Projection optimality therefore gives $\log S \in \widehat{N}_P \mathfrak{F}$. Applying the first part with $q = 1$ proves the nearest-versus-best coincidence criterion. ∎

*Proof of Lemma 4.1.* Congruence by $H^{-1/2}$ is an isometry for $d_{\mathrm{AI}}$, so it is enough to compute distances between diagonal matrices. For $D(x) = \operatorname{diag}(e^{x_1}, \ldots, e^{x_d})$,

$$d_{\mathrm{AI}}(D(x), D(y)) = \left\| \log(D(x)^{-1/2} D(y) D(x)^{-1/2}) \right\|_F = \|x - y\|_2 \,,$$

because the matrices commute. The same calculation applied to absolutely continuous paths gives the Euclidean line element in $x$-coordinates. The target $\mathcal{X}_K(H)$ is the preimage of the closed set $\mathcal{C}_K \subset \mathbb{S}^d_{++}$ under the continuous map $x \mapsto H^{-1/2} D(x) H^{-1/2}$, hence is closed. Therefore the restricted complexity is exactly the Euclidean distance from $x_0$ to this closed target set, with the usual convention that the distance to the empty set is $+\infty$. ∎

*Proof of Theorem 4.2.* Suppose $\kappa(D^{-1}H) \le K$. Let

$$\lambda_{\min} = \min_{v \neq 0} \frac{v^\top H v}{v^\top D v}.$$

Then all generalized Rayleigh quotients lie in $[\lambda_{\min}, K\lambda_{\min}]$. With $E = \lambda_{\min} D$,

$$E \preceq H \preceq KE.$$

Conversely, if $E \preceq H \preceq KE$ for a positive diagonal $E$, then every generalized Rayleigh quotient $v^\top Hv / v^\top Ev$ lies in $[1, K]$, so $\kappa(E^{-1}H) \le K$. ∎

*Proof of Corollary 4.4.* If $H$ and $G_0$ are diagonal, then $S_0 = H^{-1/2} G_0 H^{-1/2}$ is diagonal. The full SPD projection onto $\mathcal{C}_K$ keeps the eigenvectors of $S_0$ and clips only its log-spectrum. Therefore the optimal endpoint and the affine-invariant geodesic from $S_0$ to that endpoint remain diagonal. The diagonal family contains a full SPD shortest path, and Proposition 3.8 gives equality. ∎

*Proof of Theorem 4.5.* The proof is identical to Theorem 4.2, replacing the diagonal cone by the block-diagonal cone. If $\kappa(G^{-1}H) \le K$, take $E = \lambda_{\min} G$, which is block diagonal. The converse follows from the generalized Rayleigh quotient bound. ∎

*Proof of Proposition 4.6.* Assume, for contradiction, that there exists $E \in \mathcal{L}$ with

$$H - E \succeq 0, \qquad KE - H \succeq 0.$$

For $P, Q \succeq 0$,

$$0 \le \langle P, H - E\rangle + \langle Q, KE - H\rangle = \langle P - Q, H\rangle + \langle -P + KQ, E\rangle .$$

Since $E \in \mathcal{L}$ and $\Pi_{\mathcal{L}}(P - KQ) = 0$, the last term is zero. Thus any feasible $E$ would imply $\langle P - Q, H\rangle \ge 0$, contradicting the strict inequality in the certificate. ∎

*Proof of Corollary 4.7.* Let $q$ be a unit eigenvector with

$$q^\top (KTHT - H)q < 0.$$

Set

$$Q = qq^\top, \qquad P = K(Tq)(Tq)^\top = KTQT.$$

Then $P, Q \succeq 0$. Because $T$ is a constant sign on each block, $\Pi_{\mathcal{L}}(TQT - Q) = 0$, and hence

$$\Pi_{\mathcal{L}}(P - KQ) = K\Pi_{\mathcal{L}}(TQT - Q) = 0.$$

Moreover,

$$\langle P - Q, H\rangle = q^\top (KTHT - H)q < 0.$$

Proposition 4.6 proves infeasibility. ∎

*Proof of Proposition 4.8.* The block Jacobi matrix $G_{\mathrm{BJ}}$ is feasible in the block family, so the best block condition number is at most the one it achieves. If $\left\|\widetilde{H} - I\right\|_{\mathrm{op}} \le \delta < 1$, then

$$(1 - \delta)I \preceq \widetilde{H} \preceq (1 + \delta)I,$$

which gives the stated condition-number bound. ∎

*Proof of Proposition 5.3.* The restricted target

$$(Y + \mathcal{A}_{\mathrm{kron}}) \cap \mathcal{Y}_K$$

is a subset of $\mathcal{Y}_K$. Distance to a subset is at least distance to the full set. If the full projection lies in the restricted target, it attains both distances. ∎

*Proof of Proposition 5.4.* For $G = B \otimes A$,

$$G^{-1/2}\dot{G}G^{-1/2} = Y \otimes I_m + I_n \otimes X.$$

Its Frobenius norm squared is

$$m\,\|Y\|_F^2 + n\,\|X\|_F^2 + 2\,\mathrm{tr}(X)\,\mathrm{tr}(Y),$$

which is the displayed formula. For $X = \alpha I_m$ and $Y = -\alpha I_n$, the three terms sum to zero. ∎

*Proof of Proposition* 5.5. Using Kronecker identities,

$$G_0^{-1/2} G_1 G_0^{-1/2} = (B_0^{-1/2} B_1 B_0^{-1/2}) \otimes (A_0^{-1/2} A_1 A_0^{-1/2}).$$

Taking the logarithm gives

$$\log(G_0^{-1/2} G_1 G_0^{-1/2}) = L_B \otimes I_m + I_n \otimes L_A.$$

The squared Frobenius norm of this matrix is

$$m \|L_B\|_F^2 + n \|L_A\|_F^2 + 2 \operatorname{tr}(L_A) \operatorname{tr}(L_B),$$

which is the claimed formula. ∎

*Proof of Proposition* 5.6. In the fixed Kronecker eigenbasis, the eigenvalues of $X_t$ and $Y_t$ are $\dot{\alpha}_i$ and $\dot{\beta}_j$. By Proposition 5.4,

$$\left\|\dot{G}_t\right\|_{G_t,\mathrm{AI}}^2 = n \sum_i \dot{\alpha}_i^2 + m \sum_j \dot{\beta}_j^2 + 2 \left(\sum_i \dot{\alpha}_i\right) \left(\sum_j \dot{\beta}_j\right).$$

On the other hand,

$$\left\|\dot{Z}_t\right\|_F^2 = \sum_{i,j} (\dot{\alpha}_i + \dot{\beta}_j)^2,$$

which expands to the same expression. ∎

*Proof of Theorem* 5.7. If $M = B \otimes A$, then the spectral calculus for Kronecker products gives

$$\log M = (\log B) \otimes I_m + I_n \otimes (\log A) \in \mathcal{L}_{m,n}.$$

Conversely, if

$$\log M = Y \otimes I_m + I_n \otimes X,$$

then the two summands commute, and therefore

$$M = \exp(Y \otimes I_m) \exp(I_n \otimes X) = \exp(Y) \otimes \exp(X).$$

This proves the log characterization.

For the projection formula, decompose

$$\mathcal{L}_{m,n} = \operatorname{span}\{I_{mn}\} \oplus \{I_n \otimes X : \operatorname{tr} X = 0\} \oplus \{Y \otimes I_m : \operatorname{tr} Y = 0\},$$

an orthogonal direct sum under the Frobenius inner product. With

$$P = \tau I_{mn} + I_n \otimes X_0 + Y_0 \otimes I_m,$$

the definitions of $\tau, X_0, Y_0$ give

$$\operatorname{tr} P = \operatorname{tr} L, \qquad \operatorname{Tr}_B(P) = \operatorname{Tr}_B(L), \qquad \operatorname{Tr}_A(P) = \operatorname{Tr}_A(L).$$

Thus $L - P$ is orthogonal to each component of $\mathcal{L}_{m,n}$, so $P = \Pi_{\mathcal{L}} L$. The residual statement follows by applying the log characterization to $L = \log M$. ∎

*Proof of Lemma 5.8.* Closed geodesically convex subsets of a Hadamard manifold are proximinal and the projection is unique because squared distance is strictly convex along geodesics. The same CAT(0) convexity inequality gives, for any geodesic $\gamma$ in $C$,

$$f(\gamma_t) \leq (1-t)f(\gamma_0) + tf(\gamma_1) - \frac{1}{2}t(1-t)d(\gamma_0,\gamma_1)^2,$$

which is 1-strong geodesic convexity of $f(y) = d(y,x)^2/2$.

For the line-search statement, the descent curve is a geodesic initialized in the negative restricted-gradient direction. On a compact geodesic neighborhood where the restricted gradient is $L$-Lipschitz, the geodesic descent lemma gives

$$f(\gamma_\eta) \leq f(\gamma_0) - \eta \left\|\operatorname{grad}_C f(\gamma_0)\right\|^2 + \frac{L}{2}\eta^2 \left\|\operatorname{grad}_C f(\gamma_0)\right\|^2.$$

Thus all sufficiently small positive $\eta$ satisfy the Armijo inequality. Compactness makes the admissible upper step size uniform over the sublevel set, and geometric backtracking then accepts a step bounded below by a positive constant depending only on the Lipschitz constant and the backtracking parameters. ∎

*Proof of Theorem 5.9.* By Theorem 5.7, $\mathcal{K}_{m,n} = \exp(\mathcal{L}_{m,n})$. The principal matrix logarithm is a global diffeomorphism from $\mathbb{S}^{mn}_{++}$ to $\mathbb{S}^{mn}$, and $\mathcal{L}_{m,n}$ is a linear subspace. Hence $\mathcal{K}_{m,n}$ is a smooth embedded submanifold. This formulation also removes the factor-scale gauge: different pairs $(X,Y)$ that differ by $(X+\alpha I_m, Y-\alpha I_n)$ represent the same element of $\mathcal{L}_{m,n}$.

For $G_i = B_i \otimes A_i$, the affine-invariant geodesic satisfies

$$G_0^{-1/2}G_1G_0^{-1/2} = (B_0^{-1/2}B_1B_0^{-1/2}) \otimes (A_0^{-1/2}A_1A_0^{-1/2}).$$

Since $\exp(t\log(P\otimes Q)) = P^t \otimes Q^t$ for $P,Q \succ 0$, the ambient geodesic between $G_0$ and $G_1$ remains in $\mathcal{K}_{m,n}$. Hence $\mathcal{K}_{m,n}$ is totally geodesic.

It is also closed. If $B_k \otimes A_k \to G \succ 0$, fix the gauge $\det A_k = 1$. Uniform lower and upper eigenvalue bounds on $B_k \otimes A_k$ bound all eigenvalue ratios of $A_k$; the determinant gauge then bounds the eigenvalues of $A_k$ above and below, and the product eigenvalue bounds do the same for $B_k$. Passing to a subsequence gives $A_k \to A \succ 0$ and $B_k \to B \succ 0$, hence $G = B \otimes A$.

The SPD cone with $d_{\mathrm{AI}}$ is a Hadamard manifold. By Lemma 5.8, projection onto a closed geodesically convex subset is unique, giving $G_\star$. For $G = B\otimes A$, tangent vectors in whitened coordinates are exactly

$$Y \otimes I_m + I_n \otimes X \in \mathcal{L}_{m,n}.$$

Indeed, every such $Z \in \mathcal{L}_{m,n}$ generates the curve $G^{1/2}\exp(tZ)G^{1/2} \in \mathcal{K}_{m,n}$, while differentiating any smooth factor curve $B_t \otimes A_t$ and whitening by $G^{-1/2}$ gives an element of this same subspace. Thus the normal space is the Frobenius orthogonal complement of $\mathcal{L}_{m,n}$ in whitened coordinates. At the projection point, the logarithmic residual

$$R_\star = \log(G_\star^{-1/2}MG_\star^{-1/2})$$

is orthogonal to every tangent direction. Orthogonality to $I_n \otimes X$ and $Y \otimes I_m$ is exactly

$$\operatorname{Tr}_B(R_\star) = 0, \qquad \operatorname{Tr}_A(R_\star) = 0.$$

Conversely, these equations give the first-order projection condition on a closed geodesically convex set in a Hadamard manifold, hence the unique projection. The distance formula is the definition of $d_{\mathrm{AI}}$. ∎

*Proof of Corollary* 5.13. The equivalence $V(G) = 0 \iff G = P_{\mathcal{K}}(M)$ is the normal equation in Theorem 5.9. The quantitative bounds use Lemma 5.8: the objective

$$f(G) = \frac{1}{2} d_{\mathrm{AI}}(G, M)^2$$

is 1-strongly geodesically convex on the totally geodesic Hadamard submanifold $\mathcal{K}_{m,n}$. Let $G_\star = P_{\mathcal{K}}(M)$, $v = \mathrm{Log}_G(G_\star)$, and $d = \|v\| = d_{\mathrm{AI}}(G, G_\star)$. Strong convexity along the geodesic from $G$ to $G_\star$ gives

$$f(G) - f(G_\star) \le \|V(G)\|_F \, d - \frac{1}{2} d^2.$$

Strong convexity at the minimizer gives

$$f(G) - f(G_\star) \ge \frac{1}{2} d^2.$$

Combining the inequalities yields $d \le \|V(G)\|_F$, and maximizing $\|V(G)\|_F \, d - d^2/2$ over $0 \le d \le \|V(G)\|_F$ gives the objective-gap bound. ∎

*Proof of Corollary* 5.14. If $G_r = B_r \otimes A_r$, then $V_r \in \mathcal{L}_{m,n}$ has the form

$$V_r = Y_r \otimes I_m + I_n \otimes X_r.$$

The two summands commute, so $\exp(\eta V_r) = \exp(\eta Y_r) \otimes \exp(\eta X_r)$, and the trial point remains in $\mathcal{K}_{m,n}$.

Let $G_\star = P_{\mathcal{K}}(M)$. The restricted negative gradient of

$$f(G) = \frac{1}{2} d_{\mathrm{AI}}(G, M)^2$$

is represented in whitened coordinates by $V(G) = \Pi_{\mathcal{L}} R_G(M)$. The initial sublevel set $\mathcal{Q}_0 = \{G \in \mathcal{K}_{m,n} : f(G) \le f(G_0)\}$ is closed and bounded in the complete finite-dimensional Hadamard submanifold $\mathcal{K}_{m,n}$, hence compact by Hopf–Rinow. Since the SPD cone has no cut locus and the squared-distance objective is smooth, the restricted gradient is Lipschitz on a compact geodesic neighborhood of $\mathcal{Q}_0$; let $L_0$ be such a constant. By Lemma 5.8, Armijo backtracking terminates and the accepted step sizes have a positive lower bound $\eta_{\min} > 0$ depending only on $\mathcal{Q}_0, L_0, \bar{\eta}, \gamma$, and $c$ [1, 8].

Armijo decrease gives

$$f(G_{r+1}) \le f(G_r) - c\eta_r \|V_r\|_F^2.$$

Thus $f(G_r)$ decreases and $\sum_r \eta_r \|V_r\|_F^2 < \infty$. Since $\eta_r \ge \eta_{\min}$, we have $\|V_r\|_F \to 0$. Any cluster point $\bar{G}$ in the compact sublevel set satisfies $V(\bar{G}) = 0$, hence $\bar{G} = G_\star$ by Corollary 5.13. The cluster point is unique, so the full sequence converges to $G_\star$.

For the rate, set

$$\underline{\eta} = \min\{\eta_{\min}, (2c)^{-1}\}.$$

Corollary 5.13 gives

$$\|V_r\|_F^2 \ge 2\big[f(G_r) - f(G_\star)\big].$$

Combining this with Armijo decrease and $\eta_r \ge \underline{\eta}$ yields

$$f(G_{r+1}) - f(G_\star) \le (1 - 2c\underline{\eta})\big[f(G_r) - f(G_\star)\big],$$

and iteration proves the displayed bound. ∎

*Proof of Theorem 5.15.* For the self-conditioned metric target, the condition-number set $\mathcal{C}_K$ is closed and geodesically convex because the weighted matrix geometric mean satisfies

$$\lambda_{\max}(S_0 \#_t S_1) \le \lambda_{\max}(S_0)^{1-t}\lambda_{\max}(S_1)^t$$

and the analogous lower bound for $\lambda_{\min}$. Together with Theorem 5.9, this makes $\mathcal{K}_{m,n,K} = \mathcal{K}_{m,n} \cap \mathcal{C}_K$ closed and geodesically convex.

The two displayed distance bounds are the Hadamard Pythagorean inequality for projection onto $\mathcal{K}_{m,n}$, followed by the triangle inequality through $G_\star = P_{\mathcal{K}}(M)$.

It remains to prove the QP formula. For $B \otimes A$,

$$\kappa(B \otimes A) = \kappa(B)\kappa(A),$$

so the target condition is

$$(x_m - x_1) + (y_n - y_1) \le \log K$$

for sorted log-eigenvalues $x_i$ of $A$ and $y_j$ of $B$. By Proposition 5.5, the squared distance between Kronecker products is

$$n\, d_{\mathrm{AI}}(A_0, A)^2 + m\, d_{\mathrm{AI}}(B_0, B)^2 + 2(\log\det A - \log\det A_0)(\log\det B - \log\det B_0).$$

The eigenvalue lower bound for the affine-invariant metric and Hoffman–Wielandt give

$$d_{\mathrm{AI}}(A_0, A)^2 \ge \sum_i (x_i - a_i)^2, \qquad d_{\mathrm{AI}}(B_0, B)^2 \ge \sum_j (y_j - b_j)^2.$$

The determinant terms are the corresponding sums of log-eigenvalue differences. Expanding gives the lower bound

$$\sum_{i,j} (x_i + y_j - a_i - b_j)^2.$$

Equality is attained by choosing $A$ and $B$ with the same eigenvectors as $A_0$ and $B_0$, proving the convex QP. ∎

*Proof of Theorem 5.16.* For any $G \succ 0$,

$$\kappa(G^{-1}H) = \kappa(H^{-1/2}GH^{-1/2}),$$

because $G^{-1}H$ is similar to $H^{1/2}G^{-1}H^{1/2} = (H^{-1/2}GH^{-1/2})^{-1}$, and a positive-definite matrix and its inverse have the same spectral condition number. Thus $G \in \mathcal{R}_{K,\mathcal{K}}(H)$ if and only if $H^{-1/2}GH^{-1/2} \in \mathcal{C}_{K,\mathcal{K}}(H)$, proving the equivalence of the first two items.

For fixed $G \succ 0$, the condition $\kappa(G^{-1}H) \le K$ is equivalent to the existence of $\lambda > 0$ such that

$$\lambda G \preceq H \preceq K\lambda G.$$

Indeed, take

$$\lambda = \lambda_{\min}(G^{-1/2}HG^{-1/2})$$

for the forward implication, and use generalized Rayleigh quotients for the reverse implication. Setting $G = B \otimes A$ gives the Kronecker Loewner sandwich. The scalar $\lambda$ can be absorbed into $A$ or $B$, giving the scale-free form.

It remains to identify the distance. The target can be written as

$$\mathcal{R}_{K,\mathcal{K}}(H) = \mathcal{K}_{m,n} \cap H^{1/2}\mathcal{C}_K H^{1/2}.$$

The set $\mathcal{C}_K$ is closed and geodesically convex, and congruence by $H^{1/2}$ is an affine-invariant isometry, so $H^{1/2}\mathcal{C}_K H^{1/2}$ is also closed and geodesically convex. Intersecting with the closed totally geodesic submanifold $\mathcal{K}_{m,n}$ gives a closed geodesically convex subset of the Kronecker manifold. The Kronecker manifold is a complete Hadamard submanifold under the induced affine-invariant metric, so projection onto this nonempty closed geodesically convex subset exists and is unique.

Finally, congruence by $H^{-1/2}$ maps paths in $\mathcal{K}_{m,n}$ isometrically to paths in $\mathcal{S}_{\mathcal{K}}(H)$. Therefore the intrinsic distance from $S_0$ to $\mathcal{C}_{K,\mathcal{K}}(H)$ equals the affine-invariant distance from $G_0$ to $\mathcal{R}_{K,\mathcal{K}}(H)$. By Proposition 3.9, this is exactly $D_{K,\mathcal{K}}(S_0; H)$. ∎

*Proof of Corollary 5.17.* The objective $\kappa(G^{-1}H)$ is invariant under positive rescaling of $G$. Hence a minimizing sequence in $\mathcal{K}_{m,n}$ can be rescaled to satisfy $\det G = \det H$. Let

$$S = H^{-1/2} G H^{-1/2}.$$

On this determinant gauge, $\det S = 1$. Since the sequence has bounded condition number, the eigenvalues of $S$ are uniformly bounded above and below. Thus the corresponding sequence of $S$'s has a convergent subsequence, and so does the sequence $G = H^{1/2} S H^{1/2}$. The family $\mathcal{K}_{m,n}$ is closed by Theorem 5.9, so the limit remains Kronecker and attains the minimum.

The equivalence

$$\mathcal{R}_{K,\mathcal{K}}(H) \neq \emptyset \quad \Longleftrightarrow \quad K \geq K^\star_{\mathcal{K}}(H)$$

is then immediate from the definition of $K^\star_{\mathcal{K}}(H)$. ∎

*Proof of Theorem 5.10.* For

$$\phi_{\mathrm{osc}}(x) = \max_i x_i - \min_i x_i,$$

the spectral subdifferential at $\log S$ is

$$\{Q_+ - Q_- : Q_\pm \succeq 0,\ \operatorname{tr} Q_\pm = 1,\ \operatorname{range}(Q_+) \subseteq E_+,\ \operatorname{range}(Q_-) \subseteq E_-\}.$$

By Theorem 3.6, the whitened subgradient for the metric variable has the opposite sign, $Q_- - Q_+$, and global Kronecker optimality is equivalent to one such subgradient lying in the normal space of the Kronecker manifold. The whitened Kronecker tangent space is

$$\mathcal{L}_{m,n} = \{Y \otimes I_m + I_n \otimes X : X \in \mathbb{S}^m,\ Y \in \mathbb{S}^n\}.$$

Frobenius orthogonality to $I_n \otimes X$ for every $X$ is equivalent to $\mathrm{Tr}_B(Q_- - Q_+) = 0$, and orthogonality to $Y \otimes I_m$ for every $Y$ is equivalent to $\mathrm{Tr}_A(Q_- - Q_+) = 0$. This proves the necessary-and-sufficient condition. When the extreme eigenvalues are simple, the only density matrices on the two one-dimensional eigenspaces are $u_+ u_+^\top$ and $u_- u_-^\top$, giving the marginal formulation.

Suppose next that $S$ has exactly two eigenvalues. Write

$$\log S = a\Pi_+ + b\Pi_-, \qquad r_\pm = \operatorname{rank} \Pi_\pm.$$

The nearest-point normal equation gives $\log S \in \widehat{N}_P\mathcal{K}$. Because the Kronecker family is scale closed, the whitened scale direction $I$ is tangent, so $\operatorname{tr}\log S = r_+a + r_-b = 0$. Since $a > b$, there is a constant $c > 0$ such that

$$\log S = c\left(\frac{\Pi_+}{r_+} - \frac{\Pi_-}{r_-}\right).$$

Taking $Q_+ = \Pi_+/r_+$ and $Q_- = \Pi_-/r_-$ makes $Q_- - Q_+ = -c^{-1}\log S$ normal. The first part then proves global condition optimality.

For the strict counterexample, $\operatorname{tr} A = \operatorname{tr} B = 0$, and hence

$$\operatorname{Tr}_B(\log H) = \operatorname{Tr}_B(B \otimes A) = 0, \qquad \operatorname{Tr}_A(\log H) = 0.$$

The normal equations in Theorem 5.9 therefore show that $I_6$ is the unique nearest Kronecker point. The relative log-eigenvalues at the identity are the products of the diagonal entries of $B$ and $A$, so their maximum and minimum are 2 and $-2$, giving $\log \kappa(H) = 4$.

The proposed metric commutes with $H$, and

$$\log(G^{-1}H) = (B - \tfrac{1}{4}I_3) \otimes A.$$

The diagonal entries of $B - \frac{1}{4}I_3$ are $7/4, -3/4, -7/4$. Multiplication by the two diagonal entries $\pm 1$ of $A$ gives extreme relative log-eigenvalues $7/4$ and $-7/4$. Their oscillation is $7/2 < 4$, proving strict noncoincidence. □

*Proof of Corollary 5.11.* Let $P = P_{\mathcal{K}}(H)$, set $S = P^{-1/2}HP^{-1/2}$, and write $L = \log S$. The projection normal equations give

$$\operatorname{Tr}_A(L) = \operatorname{Tr}_B(L) = 0.$$

Introduce the real matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The symmetric matrices $I, X, Z$ and the skew-symmetric matrix $J$ give an orthogonal tensor basis for real symmetric $4 \times 4$ matrices: the symmetric tensors are generated by the nine products of $I, X, Z$, together with $J \otimes J$. The two zero-partial-trace conditions remove every term containing an identity factor. Hence

$$L = \sum_{r,s \in \{X,Z\}} C_{rs}\, r \otimes s + cJ \otimes J$$

for a real $2 \times 2$ coefficient matrix $C$ and a scalar $c$.

Conjugation by $O \in O(2)$ acts on the span of $X, Z$ through an orthogonal $2 \times 2$ transformation, and every such transformation is realized by a suitable $O$: planar rotations act by twice their angle on $\operatorname{span}\{X, Z\}$, and a reflection supplies the other connected component of $O(2)$. Applying the singular value decomposition of $C$, choose local orthogonal matrices $O_A, O_B$ so that conjugation by $O_B \otimes O_A$ transforms $L$ into

$$L' = aX \otimes X + bZ \otimes Z + c'J \otimes J.$$

The three displayed tensor products commute. Their common orthonormal eigenbasis is the real Bell basis

$$\psi_1 = (e_1 \otimes e_1 + e_2 \otimes e_2)/\sqrt{2}, \quad \psi_2 = (e_1 \otimes e_1 - e_2 \otimes e_2)/\sqrt{2},$$
$$\psi_3 = (e_1 \otimes e_2 + e_2 \otimes e_1)/\sqrt{2}, \quad \psi_4 = (e_1 \otimes e_2 - e_2 \otimes e_1)/\sqrt{2}.$$

Every Bell projector has both marginal density matrices equal to $I_2/2$. If $\Pi_+$ and $\Pi_-$ are the extreme spectral projectors of $L'$, then

$$Q_+ = \frac{\Pi_+}{\operatorname{rank}\Pi_+}, \qquad Q_- = \frac{\Pi_-}{\operatorname{rank}\Pi_-}$$

therefore have identical marginals on both factors. Local orthogonal conjugation preserves this equality, so the same is true for the extreme eigenspaces of $L$. Theorem 5.10 proves that $P$ is condition-optimal.

If one factor has dimension one, the Kronecker family is the full SPD cone in the other factor and separation is impossible. The only remaining nontrivial factor pair below ambient dimension six is $2 \times 2$, just handled. The existing $2 \times 3$ example attains dimension six, proving minimality. ∎

*Proof of Corollary 5.12.* The whitened metric-variable subdifferential of $G \mapsto \log \kappa(G^{-1}H)$ at $P$ is

$$\mathcal{W}(P) = \{Q_- - Q_+ : Q_\pm \in \mathcal{D}_\pm\}.$$

Restricting the objective to the Kronecker manifold projects this set onto the whitened tangent space $\mathcal{L}_{m,n}$, giving exactly $\mathcal{G}(P) = \Pi_{\mathcal{L}}\mathcal{W}(P)$. The sets $\mathcal{D}_\pm$ are compact and convex, so $\mathcal{G}(P)$ is compact and convex. Strict convexity of the squared Frobenius norm gives its unique minimum-norm element $g_\star$.

By the constrained Fermat rule, $P$ is condition-optimal if and only if $0 \in \mathcal{G}(P)$, equivalently $\mu_{\mathrm{marg}}(P) = 0$. Suppose now that $g_\star \neq 0$. The variational characterization of the Euclidean projection of zero onto $\mathcal{G}(P)$ gives

$$\langle g - g_\star, g_\star \rangle \geq 0 \qquad \text{for every } g \in \mathcal{G}(P).$$

Hence

$$\min_{g \in \mathcal{G}(P)} \langle g, g_\star \rangle = \|g_\star\|_F^2 .$$

The one-sided directional derivative of a convex function is the support function of its subdifferential. Along the whitened tangent direction $-g_\star$, it is therefore

$$\max_{g \in \mathcal{G}(P)} \langle g, -g_\star \rangle = - \|g_\star\|_F^2 .$$

Because $g_\star \in \mathcal{L}_{m,n}$, the corresponding AIRM geodesic remains in the Kronecker manifold, proving the descent identity.

For simple extremes, both $\mathcal{D}_\pm$ are singletons, which gives the displayed formula. With multiplicity, minimizing the norm of a linear image of $(Q_+, Q_-)$ subject to positive-semidefinite, trace, and support constraints is a finite-dimensional convex conic problem. ∎

*Proof of Theorem 5.18.* Let

$$\delta_\star = d_{\mathrm{AI}}(H, \mathcal{K}_{m,n}) = d_{\mathrm{AI}}(H, P).$$

Since $G \in \mathcal{K}_{m,n}$, $\delta_\star \leq \rho$. By Corollary 5.13,

$$0 \leq \frac{1}{2}\rho^2 - \frac{1}{2}\delta_\star^2 \leq \frac{1}{2}\eta^2,$$

and hence $\delta_\star \geq \sqrt{\max\{\rho^2 - \eta^2, 0\}} = \delta_-$. This proves the distance bracket.

The family $\mathcal{K}_{m,n}$ is closed under positive rescaling, so Corollary 3.7 gives

$$K^\star_{\mathcal{K}}(H) \geq \exp(\alpha_{mn}\delta_\star) \geq \exp(\alpha_{mn}\delta_-).$$

The upper bound $K^\star_{\mathcal{K}}(H) \leq \kappa_G$ holds because $G$ is an admissible Kronecker metric. If $\kappa_G \leq K$, then Proposition 5.23 applied with $G_c = G$ gives the displayed finite $D_{K,\mathcal{K}}$ upper bound.

It remains to prove the exact-projection condition bracket. Again by Corollary 5.13,

$$d_{\mathrm{AI}}(G, P) \leq \eta.$$

Therefore every log-eigenvalue of $G^{-1/2}PG^{-1/2}$ has absolute value at most $\eta$, and

$$e^{-\eta}G \preceq P \preceq e^{\eta}G.$$

For any nonzero $x$,

$$e^{-\eta}\frac{x^\top Hx}{x^\top Gx} \leq \frac{x^\top Hx}{x^\top Px} \leq e^{\eta}\frac{x^\top Hx}{x^\top Gx}.$$

Taking suprema and infima over generalized Rayleigh quotients gives

$$\lambda_{\max}(P^{-1}H) \leq e^{\eta}\lambda_{\max}(G^{-1}H), \qquad \lambda_{\min}(P^{-1}H) \geq e^{-\eta}\lambda_{\min}(G^{-1}H),$$

so $\kappa_P \leq e^{2\eta}\kappa_G$. Interchanging $G$ and $P$ gives $\kappa_G \leq e^{2\eta}\kappa_P$, equivalently $\kappa_P \geq e^{-2\eta}\kappa_G$.

Since $P$ is an admissible Kronecker metric, $K^\star_{\mathcal{K}}(H) \leq \kappa_P$. The scale-closed lower bound and the distance bracket give

$$\log K^\star_{\mathcal{K}}(H) \geq \alpha_{mn}\delta_\star \geq \alpha_{mn}\delta_-.$$

Combining this with $\log\kappa_P \leq \log\kappa_G + 2\eta$ proves the stated projection-suboptimality chain. The projection feasibility and projection-infeasibility statements follow from the two-sided condition bracket. The global impossibility certificate is the contrapositive of the threshold lower bound. ∎

*Proof of Proposition 5.19.* For

$$G = (V\operatorname{diag}(e^{b_j})V^\top) \otimes (U\operatorname{diag}(e^{a_i})U^\top),$$

the matrices $H$ and $G$ commute in the basis $V \otimes U$, and the eigenvalues of $G^{-1}H$ are

$$\exp(\ell_{ij} - a_i - b_j).$$

Therefore

$$\log\kappa(G^{-1}H) = \max_{i,j}(\ell_{ij} - a_i - b_j) - \min_{i,j}(\ell_{ij} - a_i - b_j).$$

The condition $\kappa(G^{-1}H) \leq K$ is thus equivalent to the existence of a scalar $c$ such that all residual log-eigenvalues lie in $[c, c + \log K]$, which is exactly the displayed linear feasibility problem. Minimizing the residual width over $a, b$ gives the fixed-basis threshold formula. Since $\mathcal{K}_{U,V} \subset \mathcal{K}_{m,n}$, feasibility in the subfamily is a full-family primal certificate; subfamily infeasibility gives no full-family obstruction. ∎

*Proof of Theorem 5.20.* For any $Z \in \mathbb{R}^{m\times n}$,

$$\max_{ij} Z_{ij} - \min_{ij} Z_{ij} = \max_{p,q\in\Delta_{mn}} \langle p - q, Z\rangle,$$

where $\Delta_{mn}$ is the probability simplex over entries. The set $\{p - q : p, q \in \Delta_{mn}\}$ is exactly

$$\mathcal{C} = \{W : \sum_{ij} W_{ij} = 0,\ \sum_{ij} |W_{ij}| \le 2\}.$$

Let

$$\mathcal{A} = \{A \in \mathbb{R}^{m\times n} : A_{ij} = a_i + b_j\}.$$

Then

$$\tau^\star_{U,V}(H) = \min_{A\in\mathcal{A}} \max_{W\in\mathcal{C}} \langle W, L - A\rangle .$$

This is a finite-dimensional linear program. Its dual is

$$\max_{W\in\mathcal{C}\cap\mathcal{A}^\perp} \langle W, L\rangle ,$$

with no duality gap. The orthogonality condition is computed from

$$\langle W, A\rangle = \sum_i a_i \sum_j W_{ij} + \sum_j b_j \sum_i W_{ij}.$$

Thus $W \in \mathcal{A}^\perp$ exactly when every row sum and every column sum vanishes. These constraints imply $\sum_{ij} W_{ij} = 0$, so the dual feasible set is the one displayed in the theorem.

If a feasible $W$ has $\langle W, L\rangle > \log K$, then $\tau^\star_{U,V}(H) > \log K$, so the fixed-basis log-width target is infeasible by Proposition 5.19. For any nonzero zero-total $W$, the identity $\sum W^+ = \sum W^- = \frac{1}{2}\|W\|_1$ gives the normalized witness form when $\|W\|_1 = 2$. ∎

*Proof of Theorem 5.21.* Proposition 5.19 proves soundness of the fixed-basis feasible status. Theorem 5.20 proves the obstruction status. The four full noncommuting statuses follow from Theorem 5.18. If none of those conditions holds, the procedure records only INCONCLUSIVE, which makes no claim. ∎

*Proof of Corollary 5.22.* The enclosure assumptions imply

$$\delta_- = \sqrt{\max\{\rho^2 - \eta^2, 0\}} \ge \sqrt{\max\{\underline{\rho}^2 - \overline{\eta}^2, 0\}} = \underline{\delta}.$$

Therefore $K^\star_{\mathcal{K}}(H) \ge \exp(\alpha_{mn}\underline{\delta})$, proving the first implication. The second follows from $\kappa_G \le \overline{\kappa}$. For the exact projection, the bracket in Theorem 5.18 and the enclosures give

$$e^{-2\overline{\eta}}\underline{\kappa} \le e^{-2\eta}\kappa_G \le \kappa_P \le e^{2\eta}\kappa_G \le e^{2\overline{\eta}}\overline{\kappa}.$$

The remaining two implications follow immediately. Returning INCONCLUSIVE when no strict certified comparison holds makes no claim and is therefore sound. ∎

*Proof of Proposition 5.23.* The matrix

$$S_c^{-1} = H^{1/2} G_c^{-1} H^{1/2}$$

has the same eigenvalues as $G_c^{-1}H$, so $\kappa(S_c) = \kappa(G_c^{-1}H)$. Hence the displayed condition is exactly membership of $S_c$ in $\mathcal{C}_K$, and $S_c \in \mathcal{C}_{K,\mathcal{K}}(H)$.

By Theorem 5.9, $\mathcal{K}_{m,n}$ is totally geodesic in the ambient affine-invariant SPD cone. Therefore the affine-invariant geodesic from $G_0$ to $G_c$ remains in $\mathcal{K}_{m,n}$ and has length

$d_{\mathrm{AI}}(G_0, G_c)$. Congruence by $H^{-1/2}$ is an isometry for the affine-invariant metric, so the relative path

$$S_t = H^{-1/2} G_t H^{-1/2}$$

is an admissible path in $\mathcal{S}_{\mathcal{K}}(H)$ from $S_0$ to the target point $S_c$ with the same length. Taking the infimum over all admissible paths gives

$$D_{K,\mathcal{K}}(S_0; H) \le d_{\mathrm{AI}}(G_0, G_c).$$

The closed-form expression for the right-hand side is Proposition 5.5. The last statement is only the logical one-way nature of a sufficient certificate. ∎

*Proof of Proposition 5.24.* The objective gradient at $Q$, in whitened coordinates and restricted to $\mathcal{L}_{m,n}$, is $-\Pi_{\mathcal{L}} R_Q(M)$. For the path

$$Q(t) = Q^{1/2} \exp(tZ) Q^{1/2},$$

the directional derivative of $\log \lambda_{\max}$ is

$$\max_{\substack{u \in E_{\max} \\ \|u\|=1}} u^\top Z u,$$

and the directional derivative of $\log \lambda_{\min}$ is

$$\min_{\substack{v \in E_{\min} \\ \|v\|=1}} v^\top Z v.$$

Thus the directional derivative of $\omega = \log \lambda_{\max} - \log \lambda_{\min}$ is the support function of the stated subdifferential. Since $K > 1$ has the strict feasible point $I_{mn}$, the convex KKT condition on the tangent Hadamard submanifold gives

$$0 \in -\Pi_{\mathcal{L}} R_\star + \mu \Pi_{\mathcal{L}} \partial\omega(Q_\star), \qquad \mu \ge 0,$$

with complementarity. This is the displayed equation; inactive constraints have $\mu = 0$. ∎

*Proof of Proposition 5.25.* Projection onto the closed geodesically convex family $\mathcal{K}_{m,n}$ satisfies the Hadamard Pythagorean inequality. Applying it to

$$H_t, \qquad P_t = P_{\mathcal{K}}(H_t), \qquad G_t \in \mathcal{K}_{m,n}$$

gives $E_t^2 \ge M_t^2 + A_t^2$. Applying Theorem 5.15 with $M = H_t$ and $G_\star = P_t$ gives the two-sided self-conditioned $K$-target bound. The computability statements are exactly Theorems 5.9 and 5.15 and Proposition 5.5. ∎

*Proof of Proposition 6.1.* From

$$(1-\varepsilon)H \preceq \widehat{H} \preceq (1+\varepsilon)H$$

we obtain

$$\frac{1}{1+\varepsilon}\frac{v^\top \widehat{H} v}{v^\top G v} \le \frac{v^\top H v}{v^\top G v} \le \frac{1}{1-\varepsilon}\frac{v^\top \widehat{H} v}{v^\top G v}.$$

Taking minima and maxima over nonzero $v$ gives the bound on generalized condition numbers. ∎

*Proof of Proposition 6.2.* On the stated Loewner event, Proposition 6.1 gives

$$\kappa(G^{-1}H) \le \frac{1+\varepsilon}{1-\varepsilon}\kappa(G^{-1}\widehat{H}) \le \frac{1+\varepsilon}{1-\varepsilon}\widehat{K} = K.$$

The event has probability at least $1-\delta$, which proves the claim. ∎

**Appendix B. Extensions.**

**B.1. Low-rank spectral monotonicity.** Let

$$\mathcal{Y}_{K,r}(y) = \{z \in \mathcal{Y}_K : \exists c \in \mathbb{R},\ |\{i : z_i \neq y_i + c\}| \leq r\}.$$

Then

$$D^{\mathrm{spec}}_{K,r}(y) = \mathrm{dist}_{\ell^2}(y, \mathcal{Y}_{K,r}(y)).$$

*Proof of Proposition 5.26.* If $r_1 \leq r_2$, then

$$\mathcal{Y}_{K,r_1}(y) \subseteq \mathcal{Y}_{K,r_2}(y) \subseteq \mathcal{Y}_K,$$

and therefore

$$D^{\mathrm{spec}}_{K,r_1}(y) \geq D^{\mathrm{spec}}_{K,r_2}(y) \geq D_K(y).$$

When $r \geq d$, the low-rank restriction disappears and equality with $D_K(y)$ holds. □